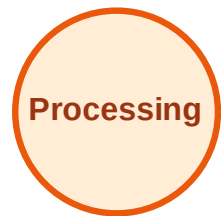


BFS Traversal

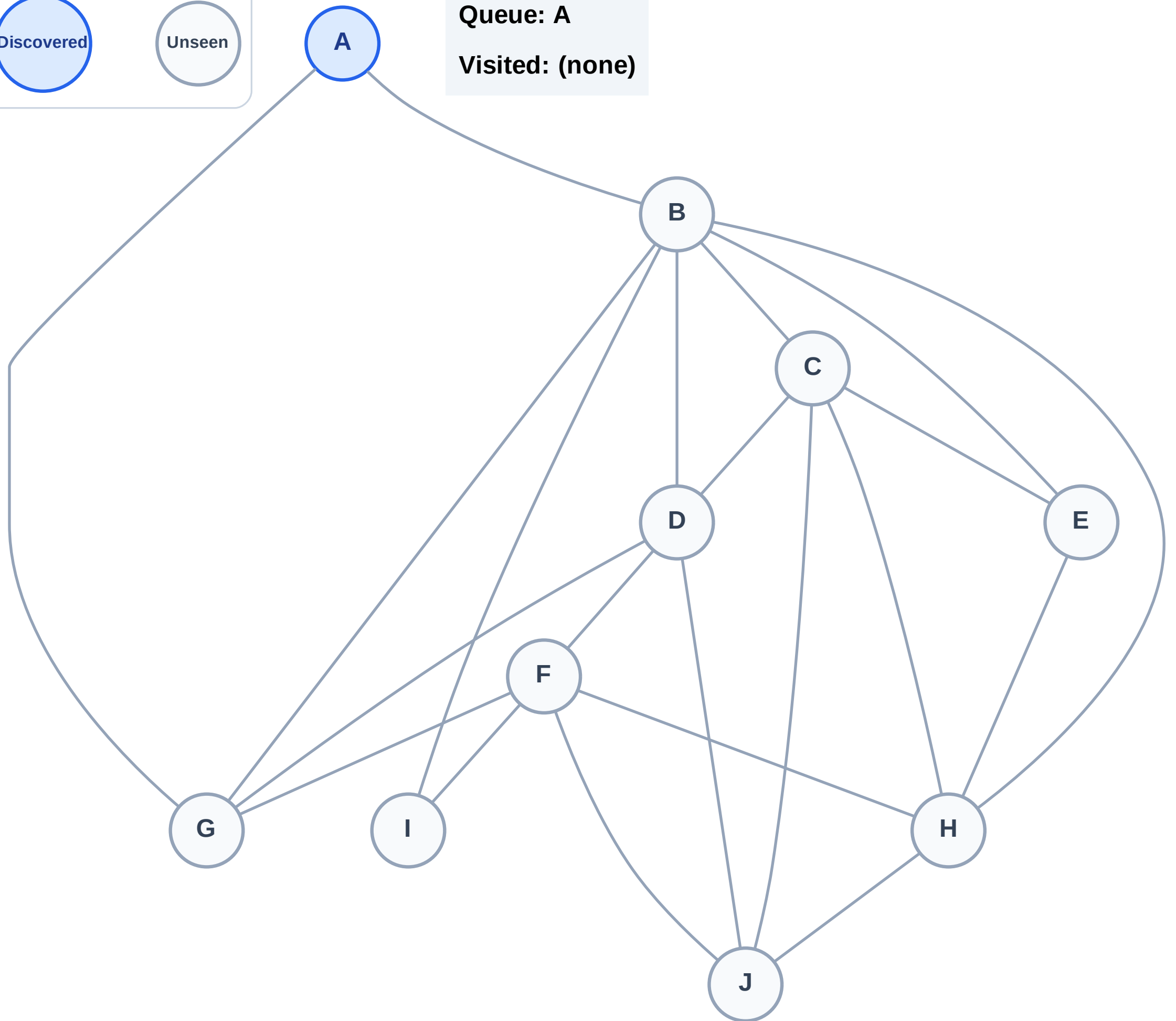
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

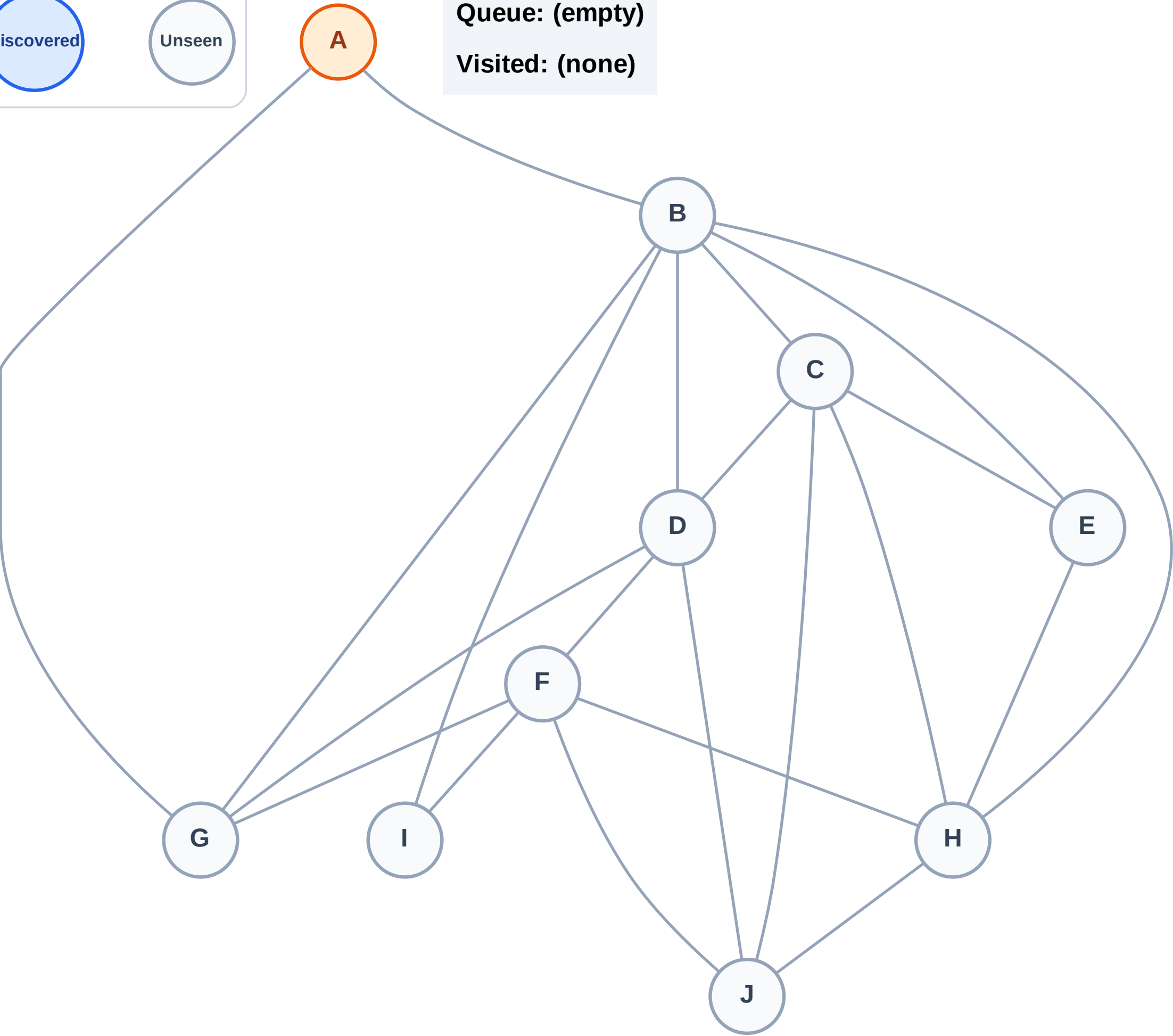
Complete

Processing

Discovered

Unseen

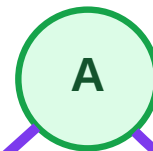
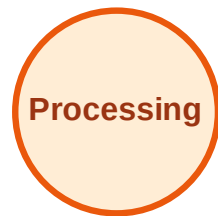
Queue: (empty)
Visited: (none)



BFS Traversal

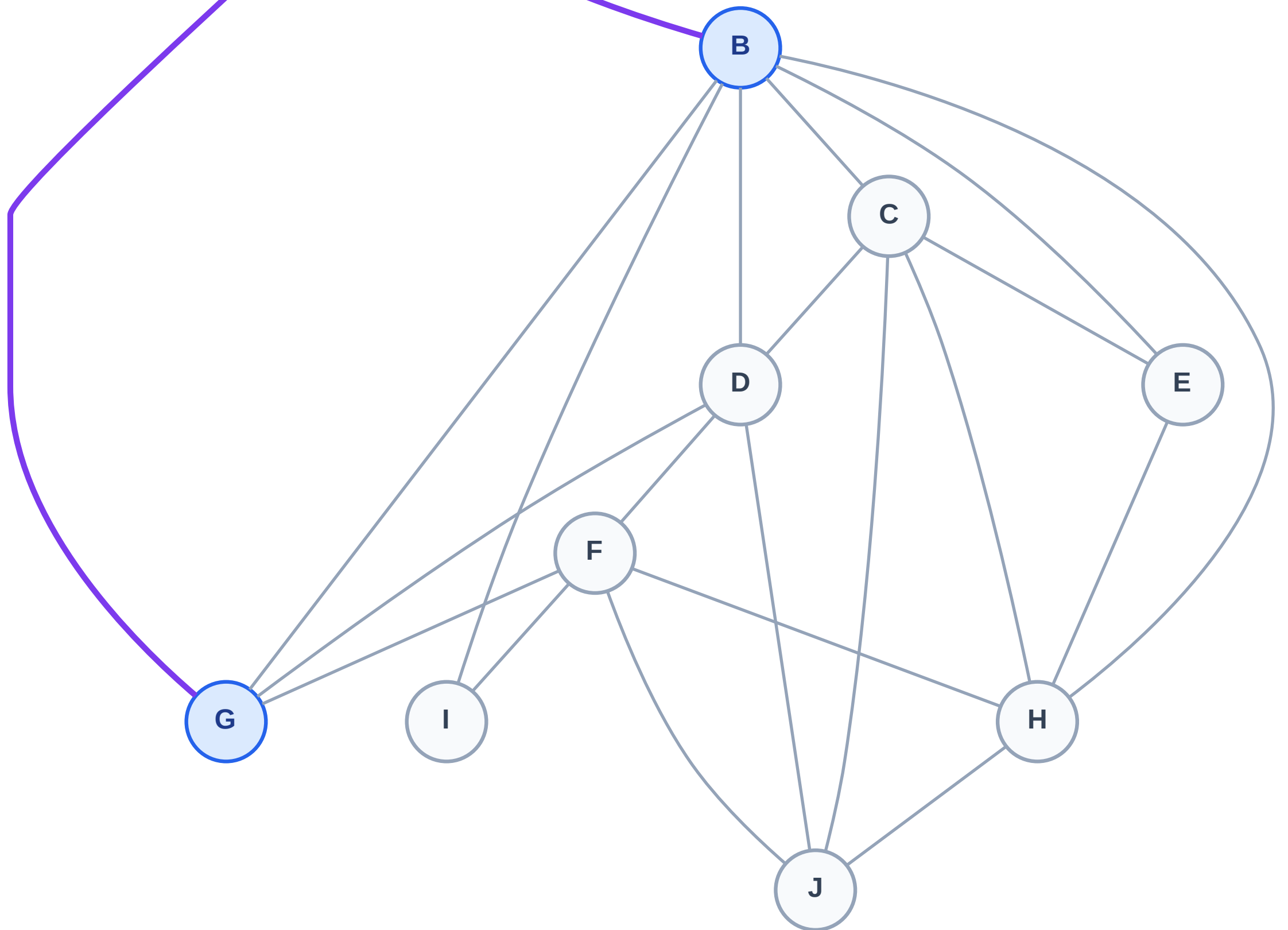
Step 3: Discover B, G from A

Legend



Queue: B → G

Visited: A



BFS Traversal

Step 4: Process node B

Legend

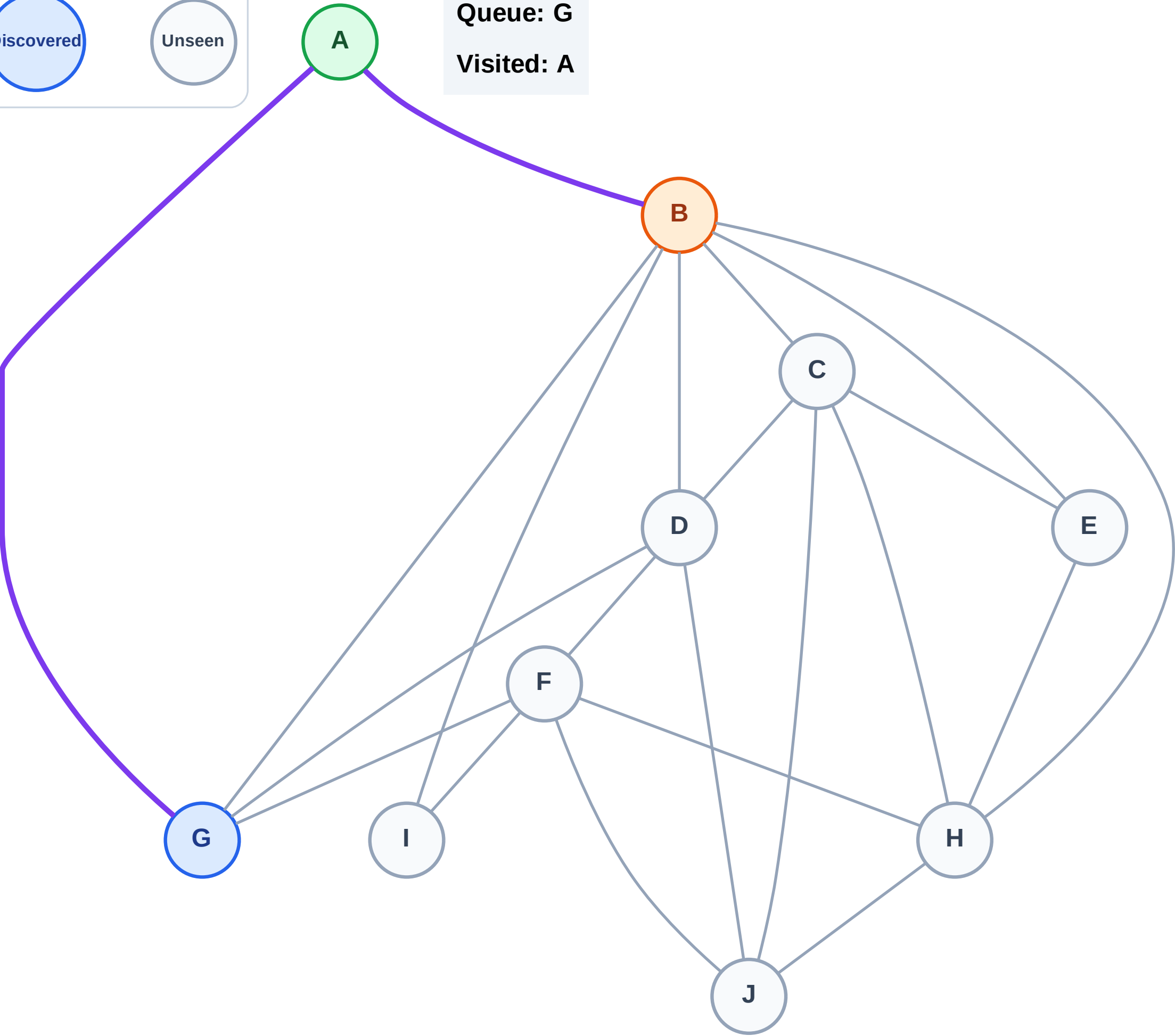
Complete

Processing

Discovered

Unseen

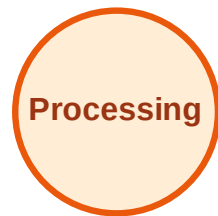
Queue: G
Visited: A



BFS Traversal

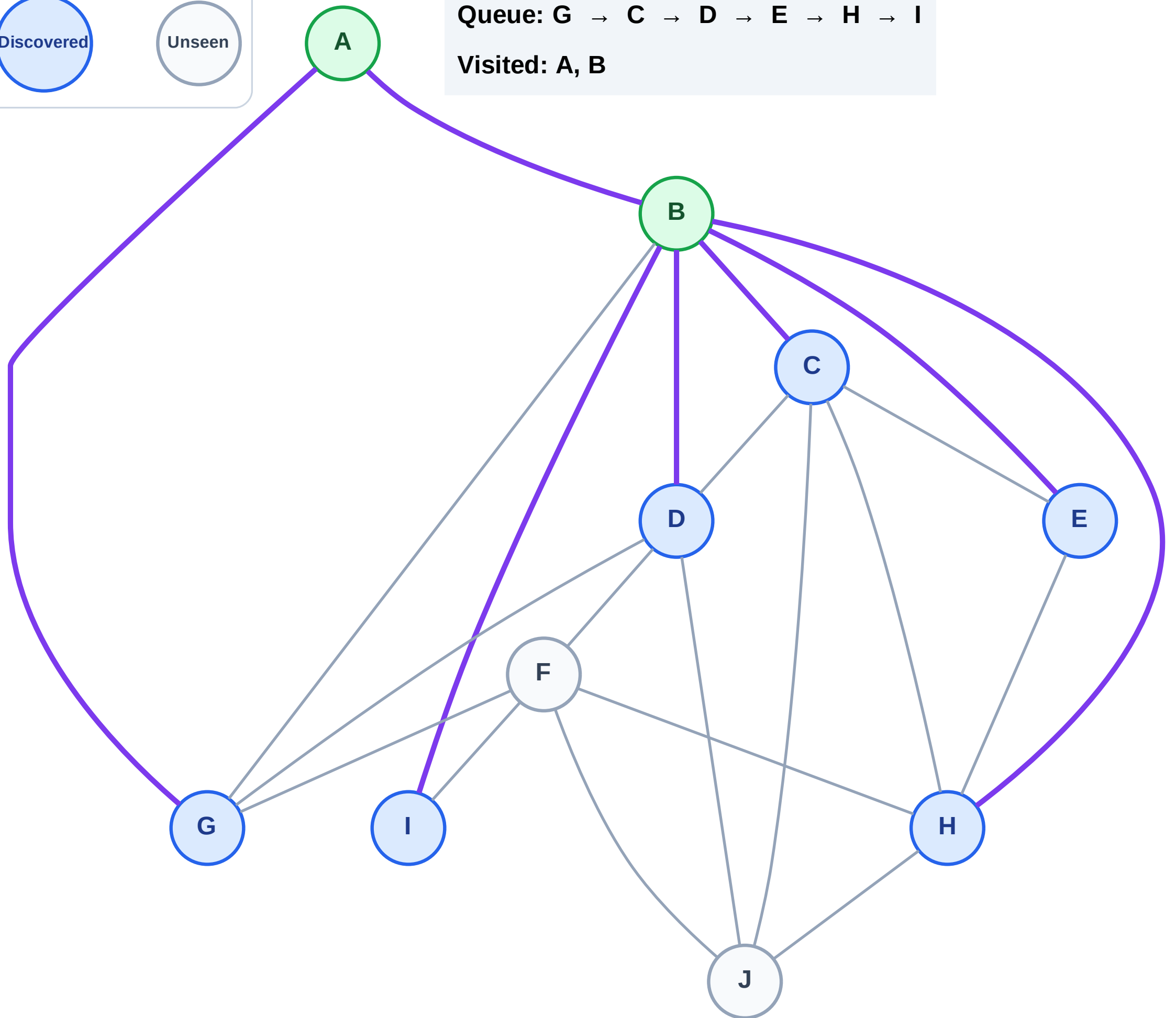
Step 5: Discover C, D, E, H, I from B

Legend



Queue: G → C → D → E → H → I

Visited: A, B



BFS Traversal

Step 6: Process node G

Legend

Complete

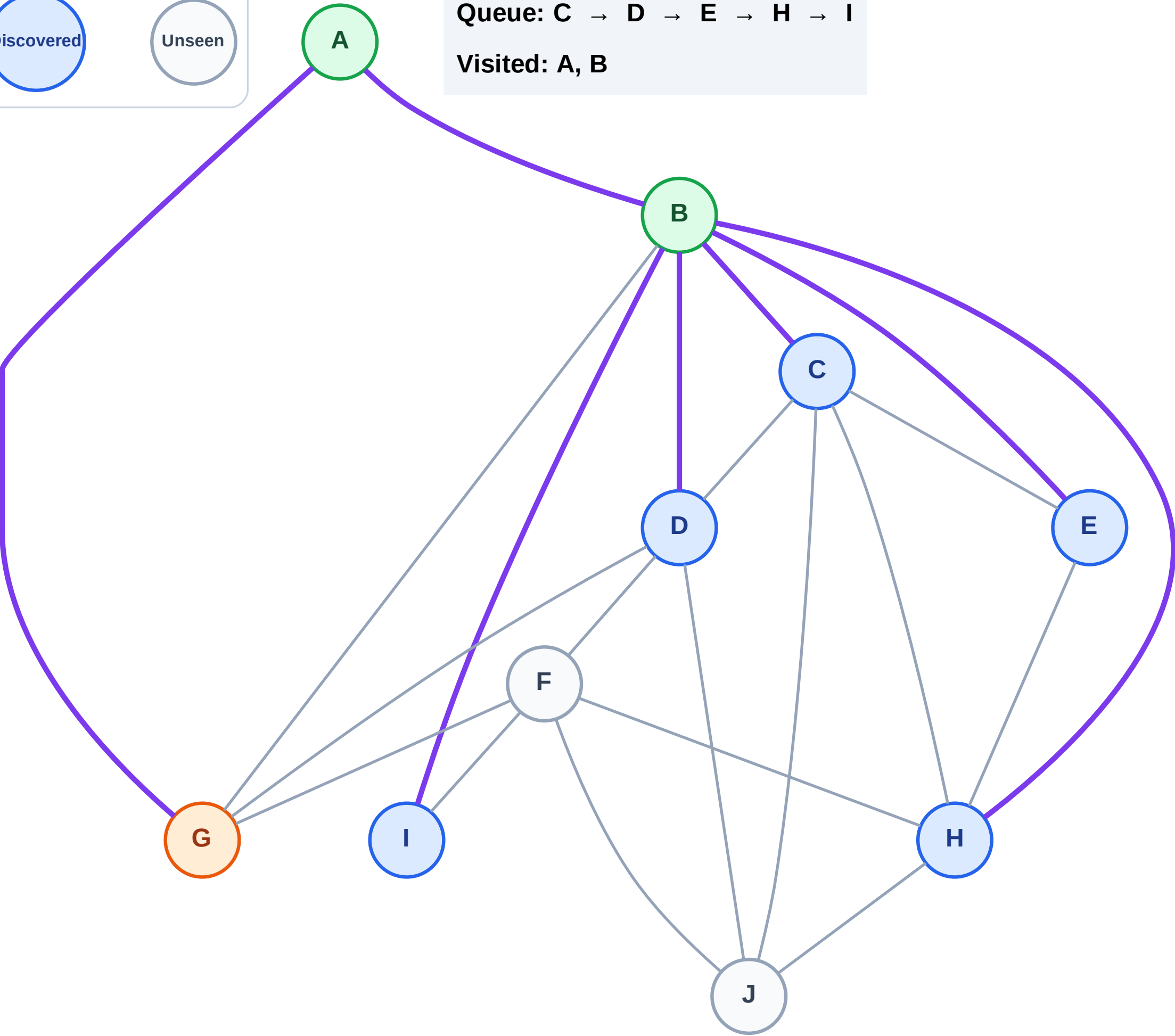
Processing

Discovered

Unseen

Queue: C → D → E → H → I

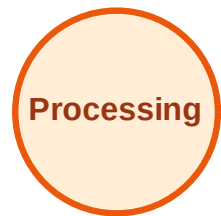
Visited: A, B



BFS Traversal

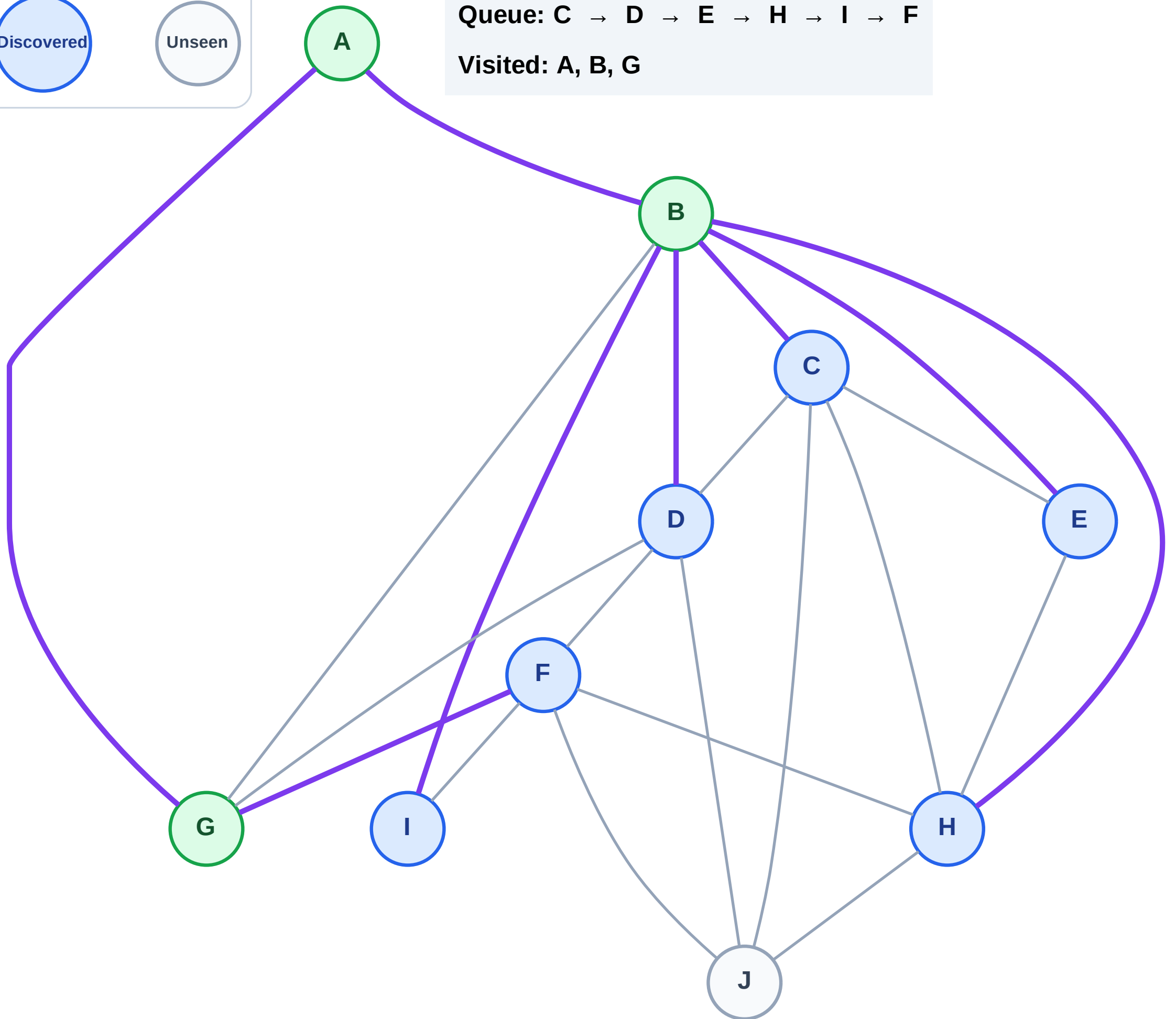
Step 7: Discover F from G

Legend



Queue: C → D → E → H → I → F

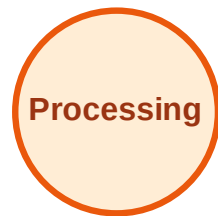
Visited: A, B, G



BFS Traversal

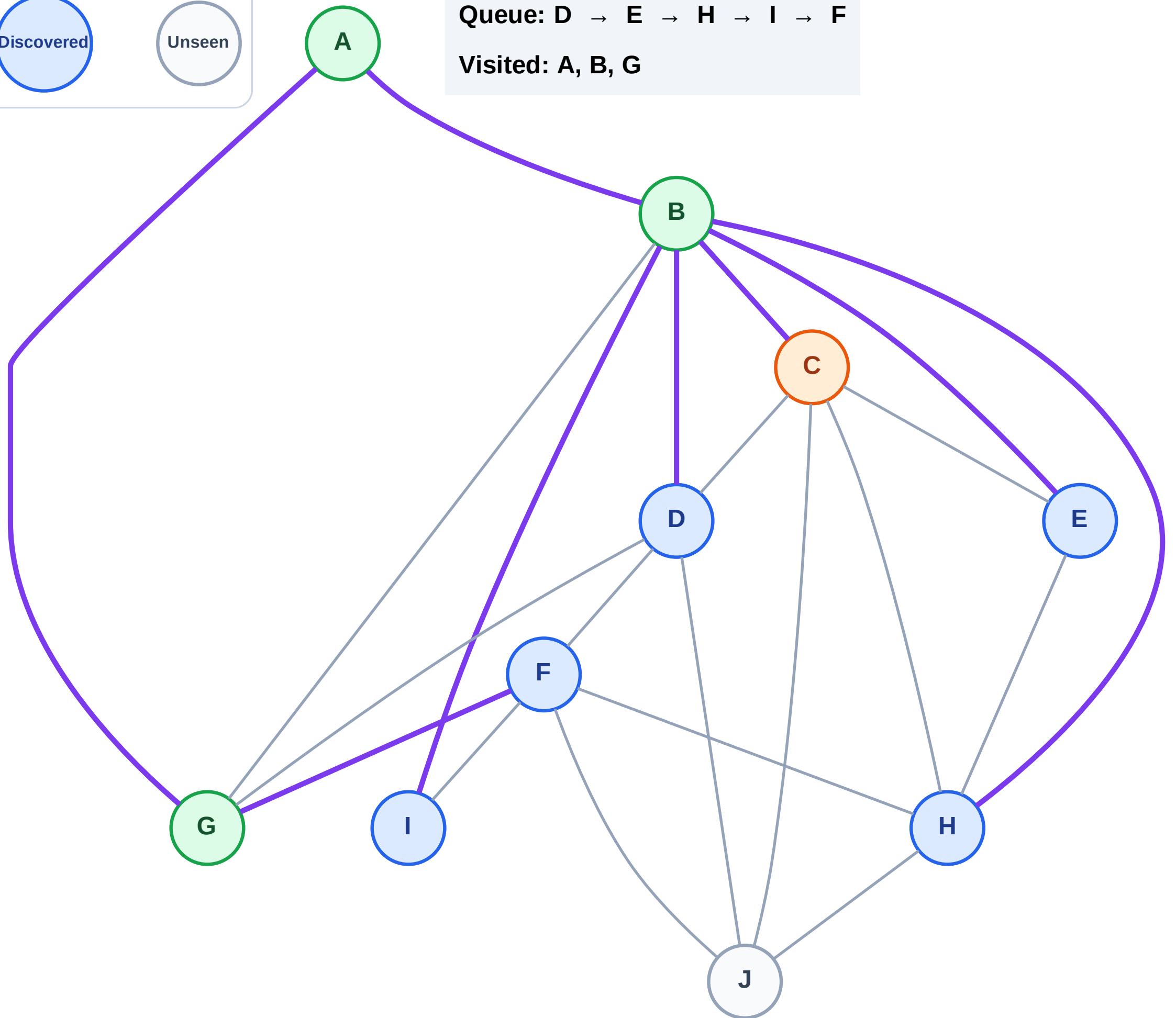
Step 8: Process node C

Legend



Queue: D → E → H → I → F

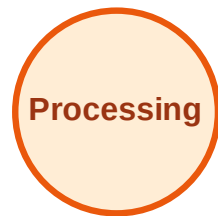
Visited: A, B, G



BFS Traversal

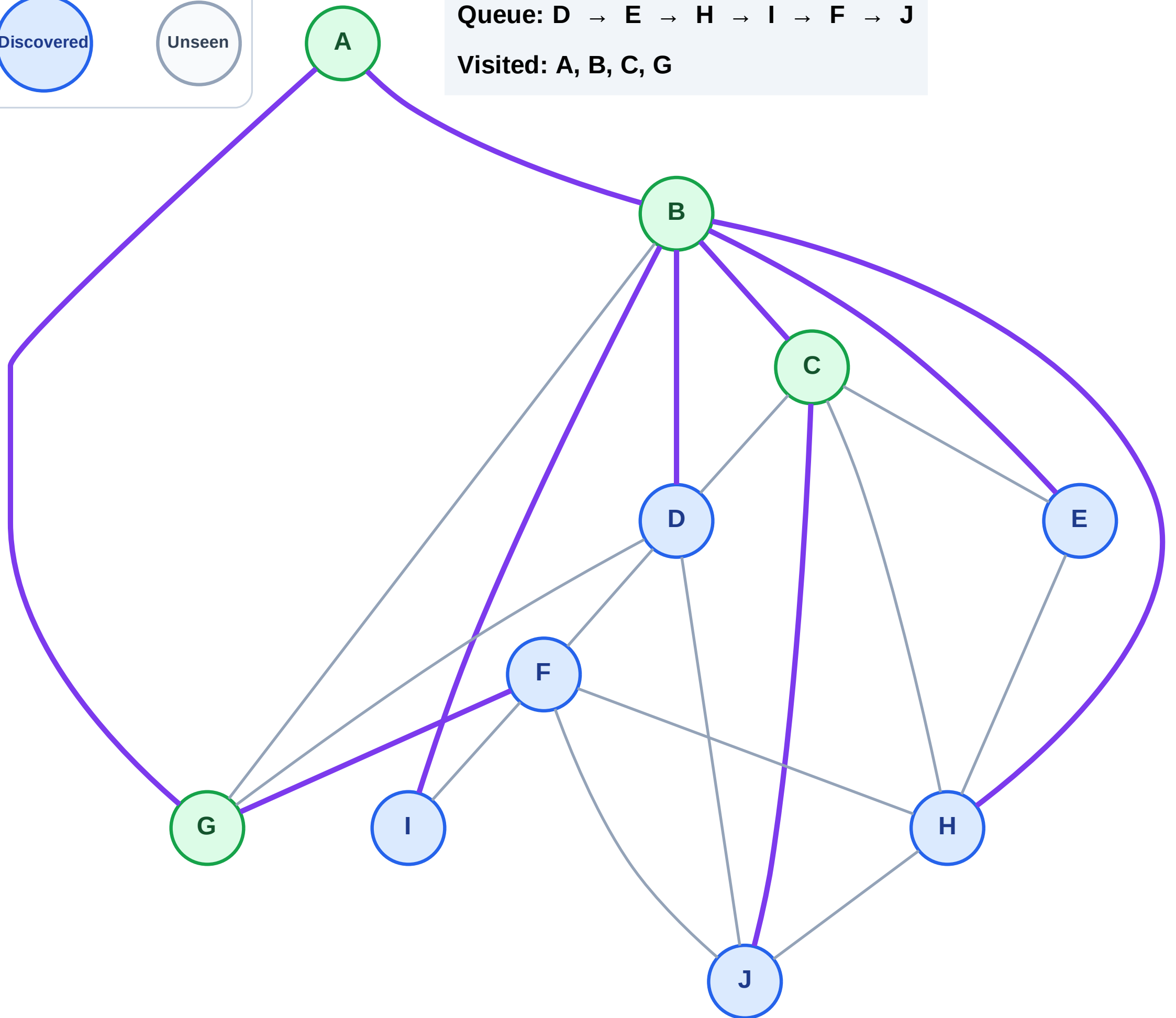
Step 9: Discover J from C

Legend



Queue: D → E → H → I → F → J

Visited: A, B, C, G



BFS Traversal

Step 10: Process node D

Legend

Complete

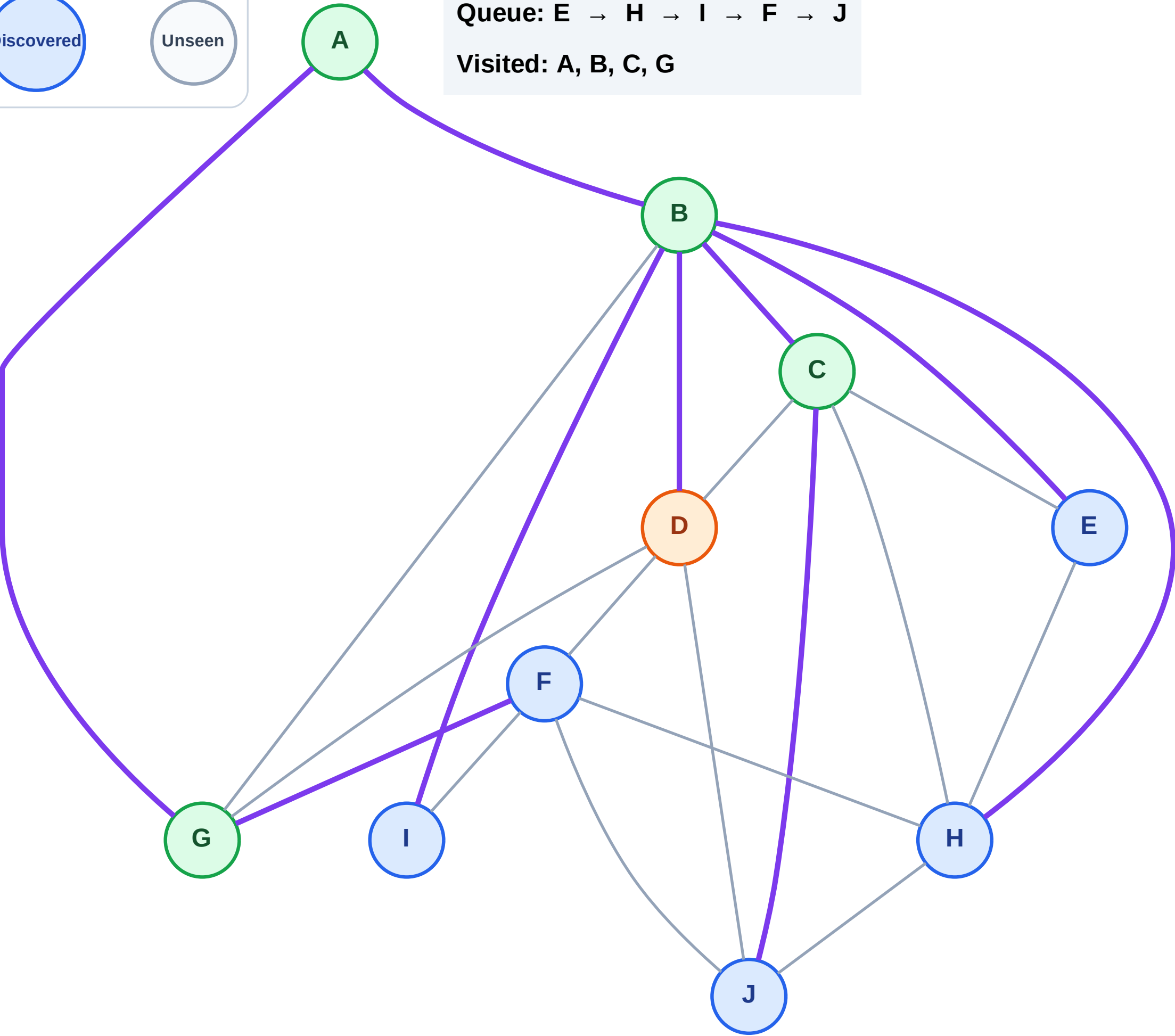
Processing

Discovered

Unseen

Queue: E → H → I → F → J

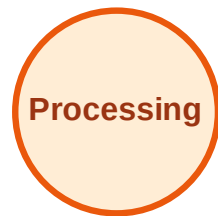
Visited: A, B, C, G



BFS Traversal

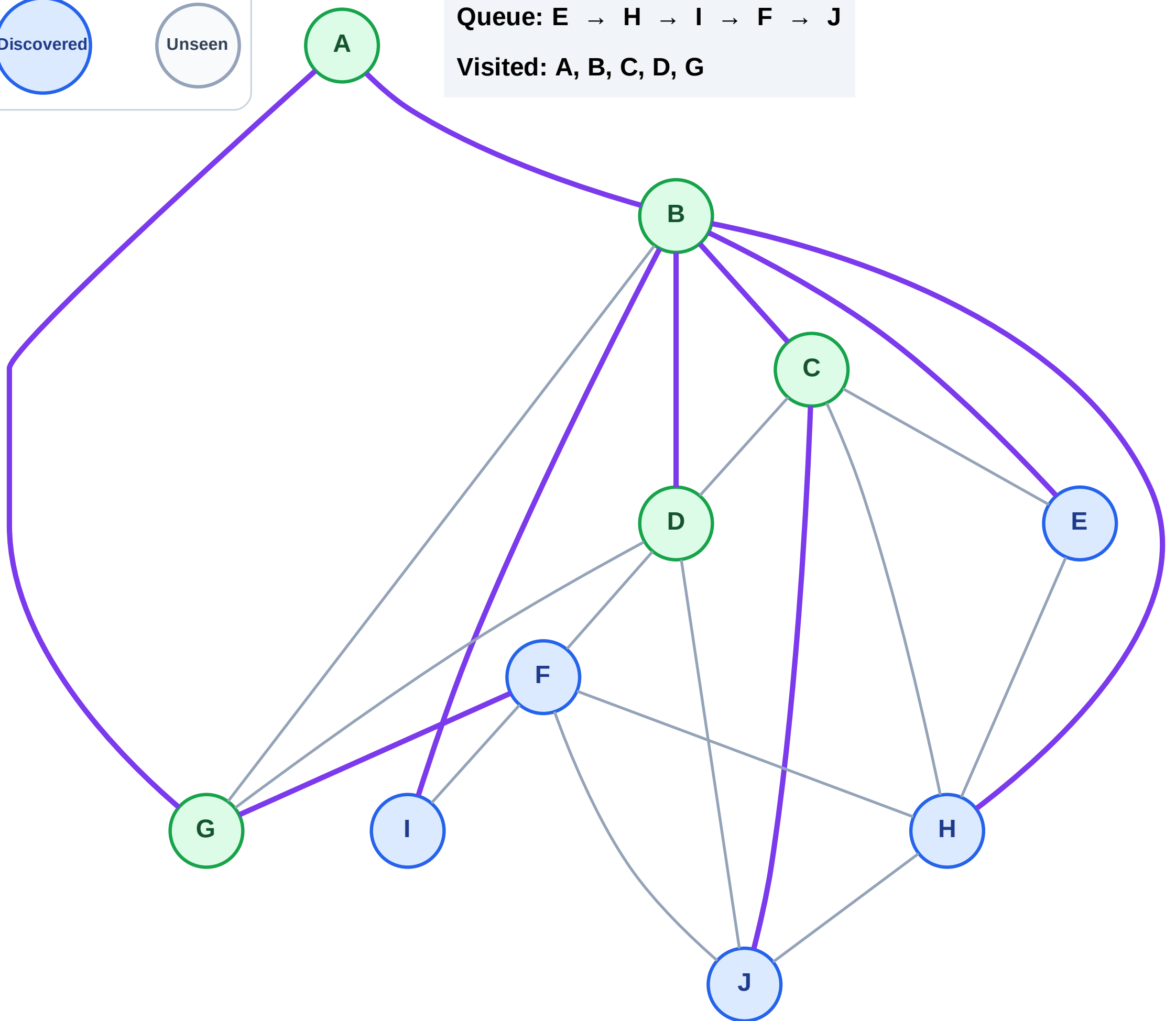
Step 11: No new neighbors from D

Legend



Queue: E → H → I → F → J

Visited: A, B, C, D, G



BFS Traversal

Step 12: Process node E

Legend

Complete

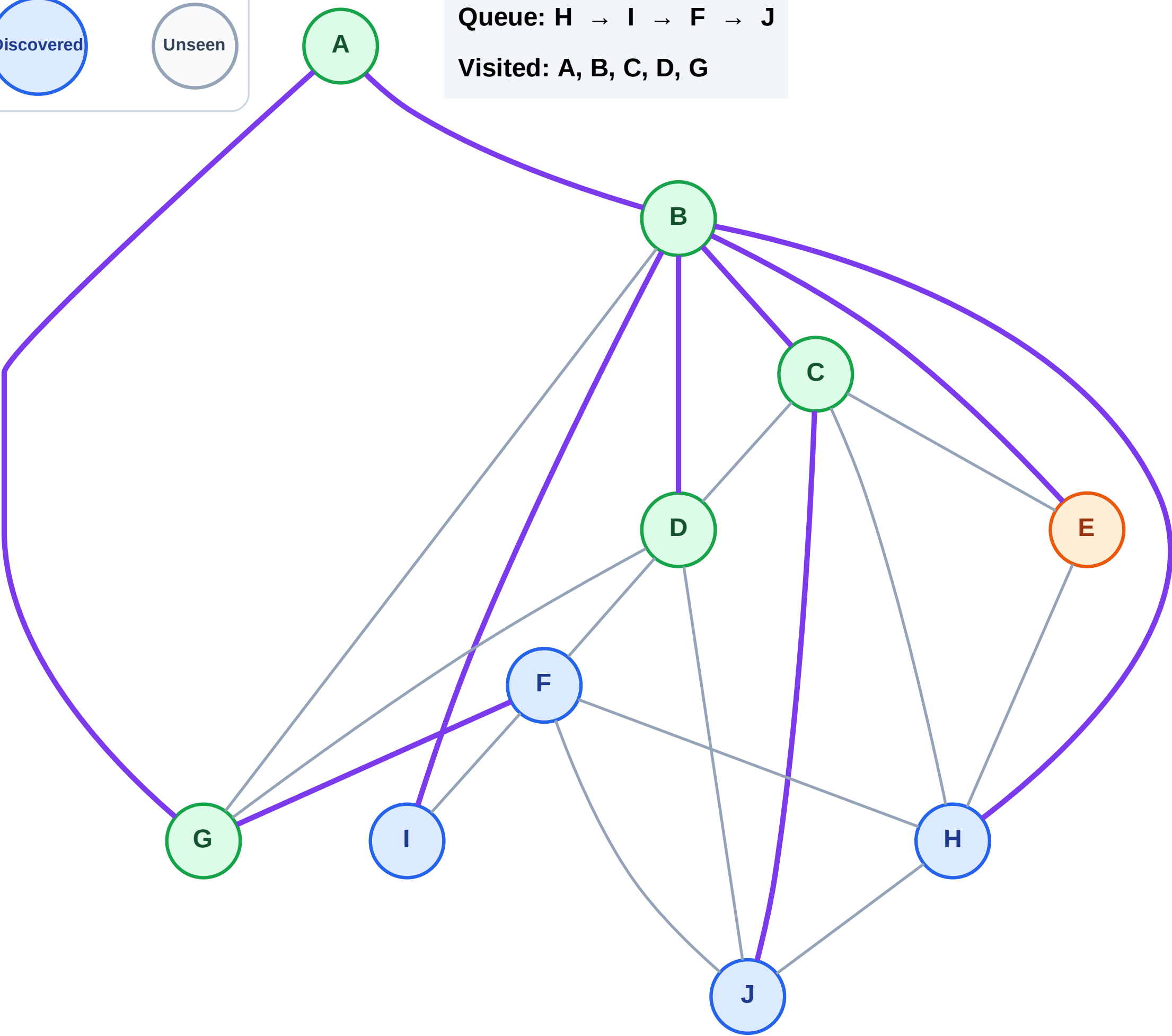
Processing

Discovered

Unseen

Queue: H → I → F → J

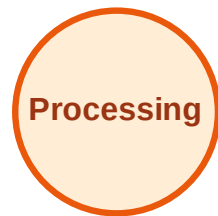
Visited: A, B, C, D, G



BFS Traversal

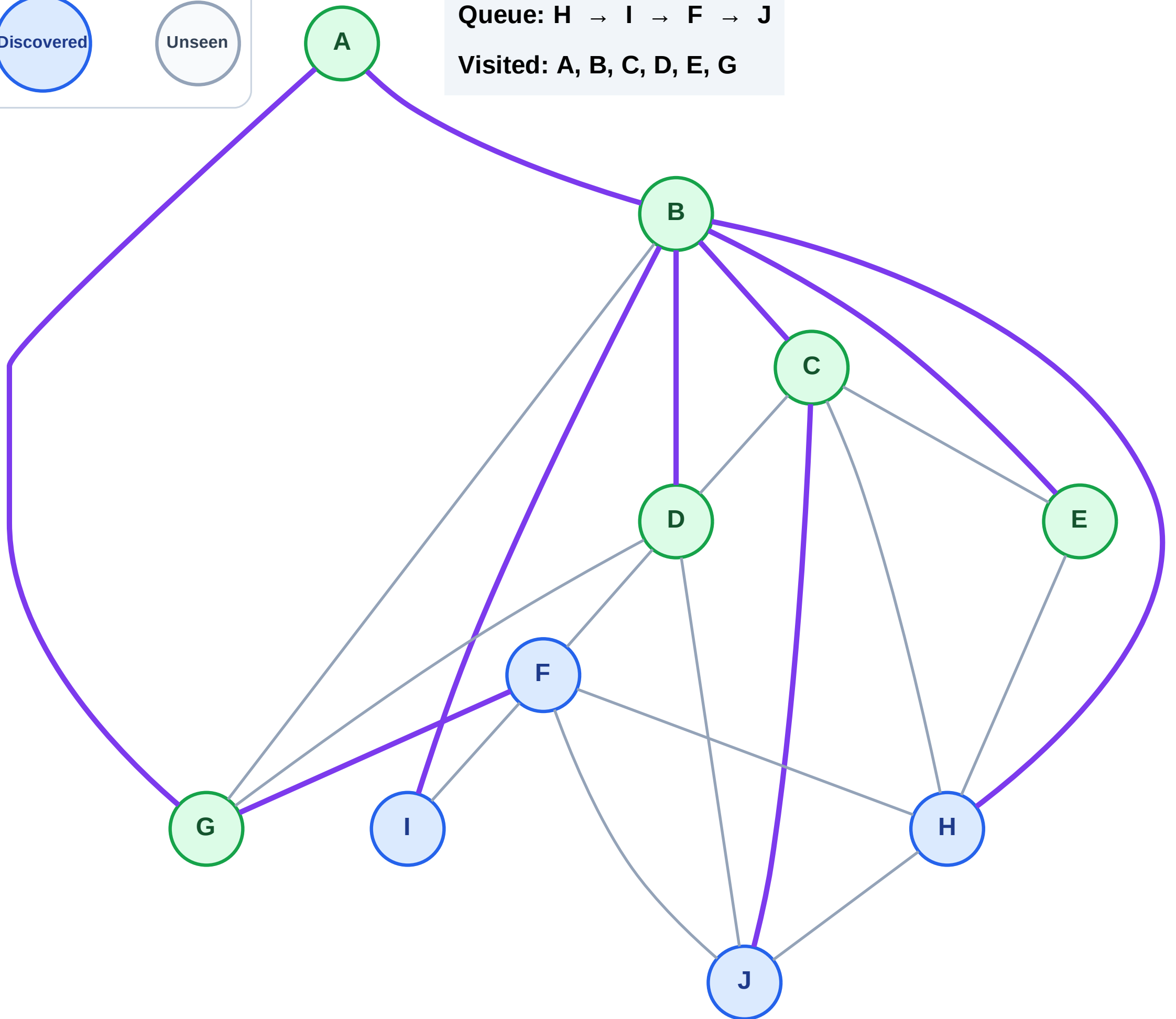
Step 13: No new neighbors from E

Legend



Queue: H → I → F → J

Visited: A, B, C, D, E, G



BFS Traversal

Step 14: Process node H

Legend

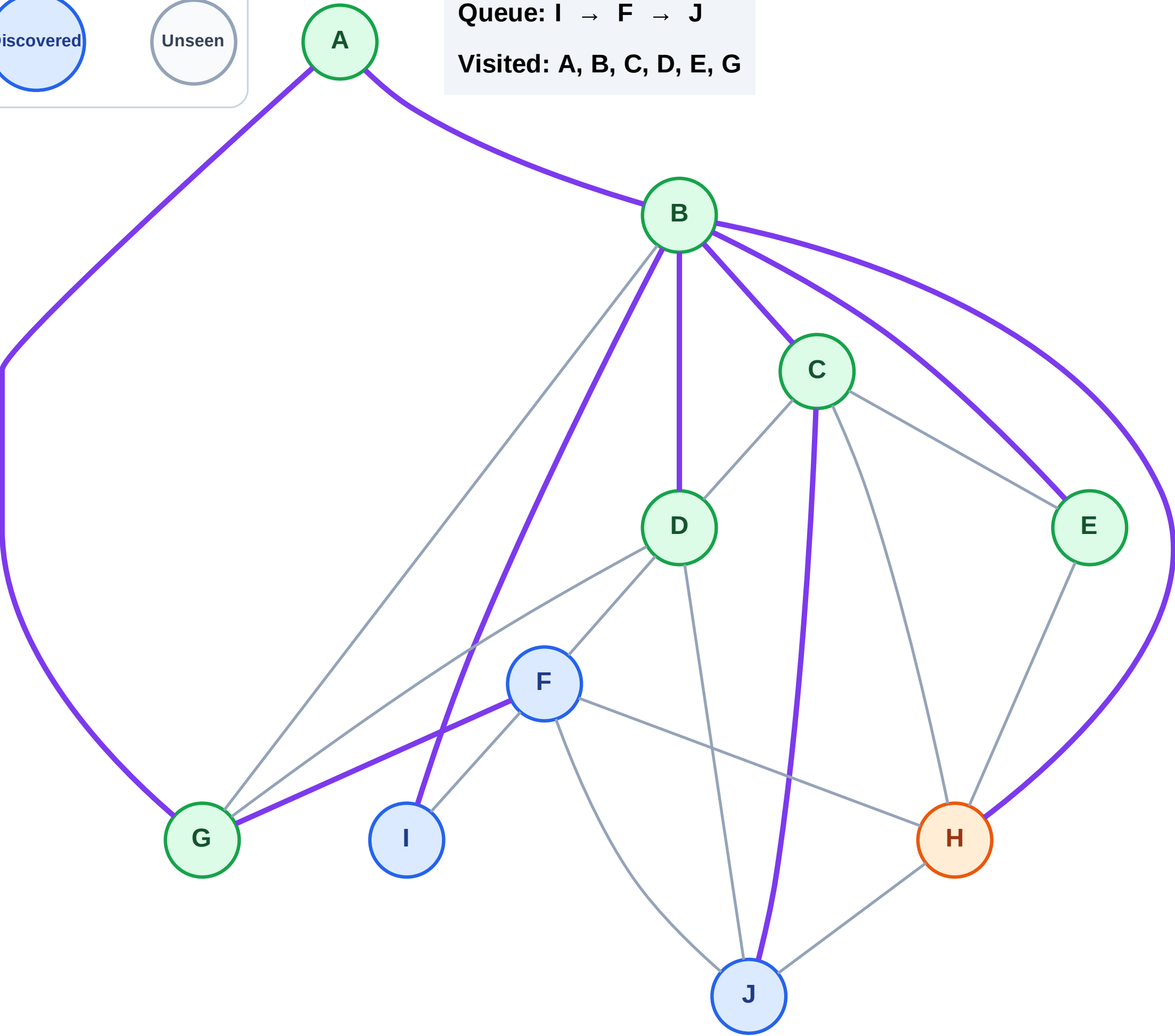
Complete

Processing

Discovered

Unseen

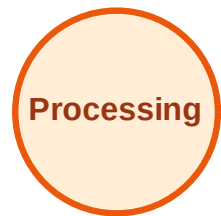
Queue: I → F → J
Visited: A, B, C, D, E, G



BFS Traversal

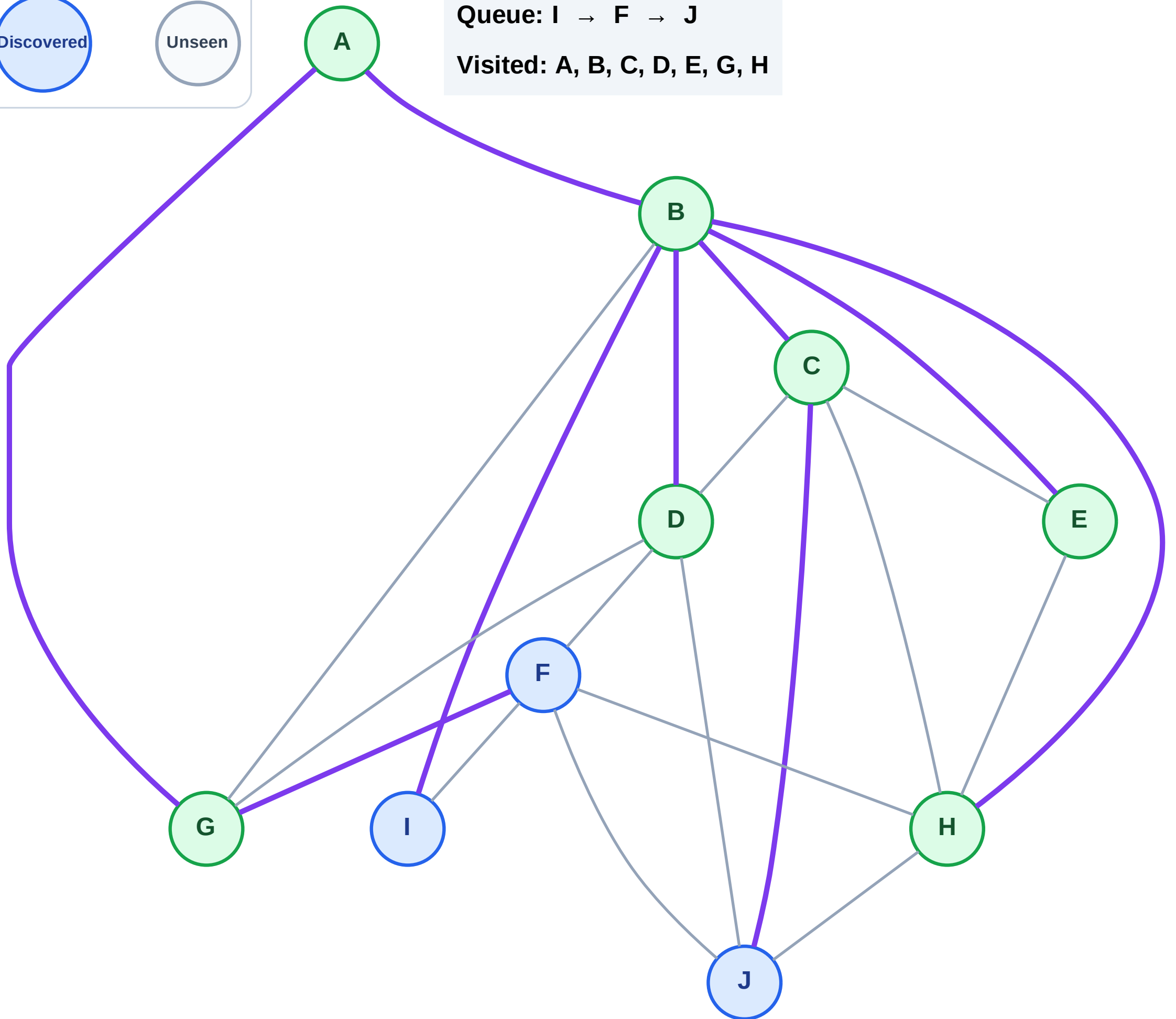
Step 15: No new neighbors from H

Legend



Queue: I → F → J

Visited: A, B, C, D, E, G, H



BFS Traversal

Step 16: Process node I

Legend

Complete

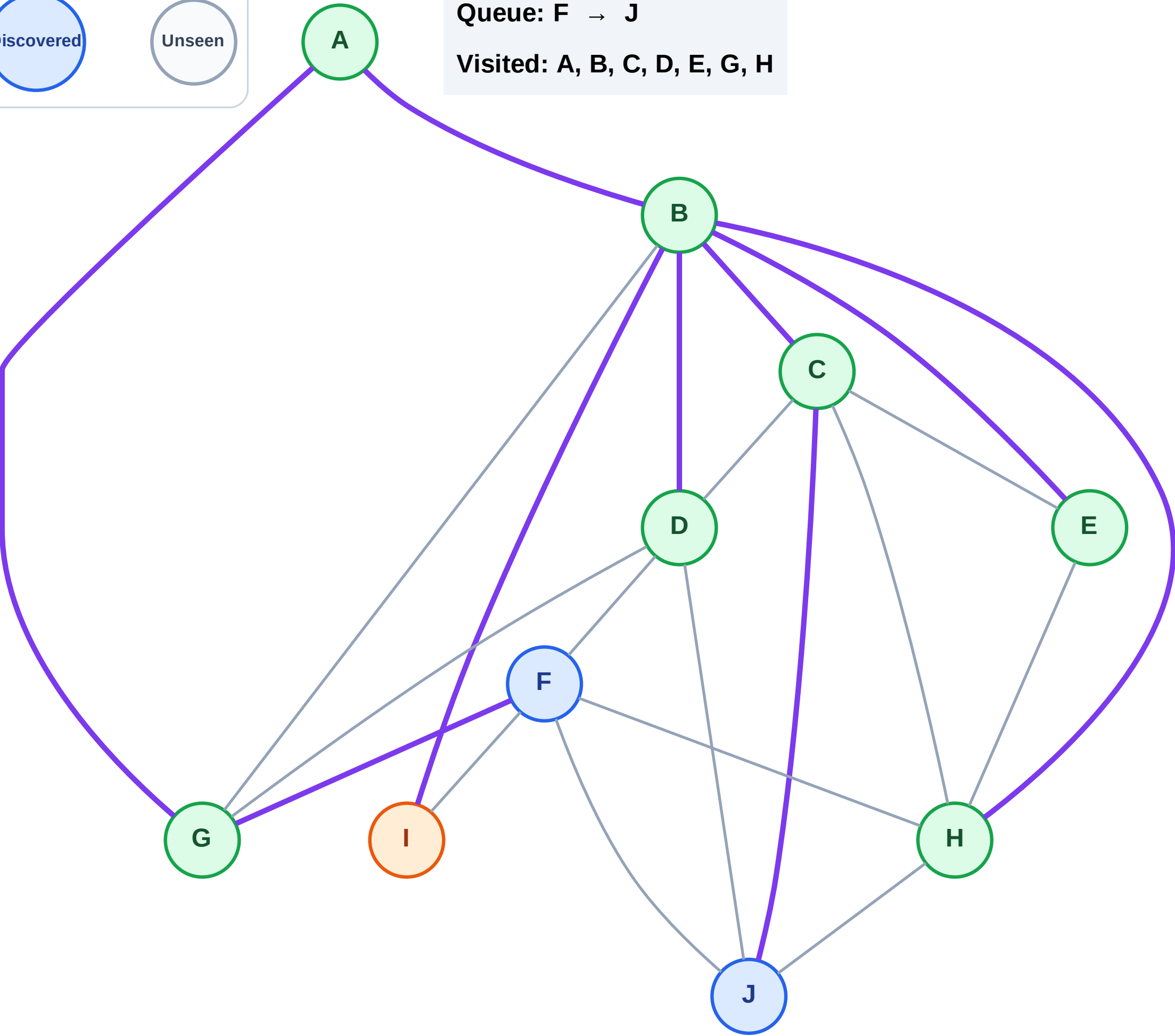
Processing

Discovered

Unseen

Queue: F → J

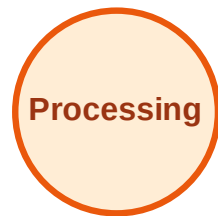
Visited: A, B, C, D, E, G, H



BFS Traversal

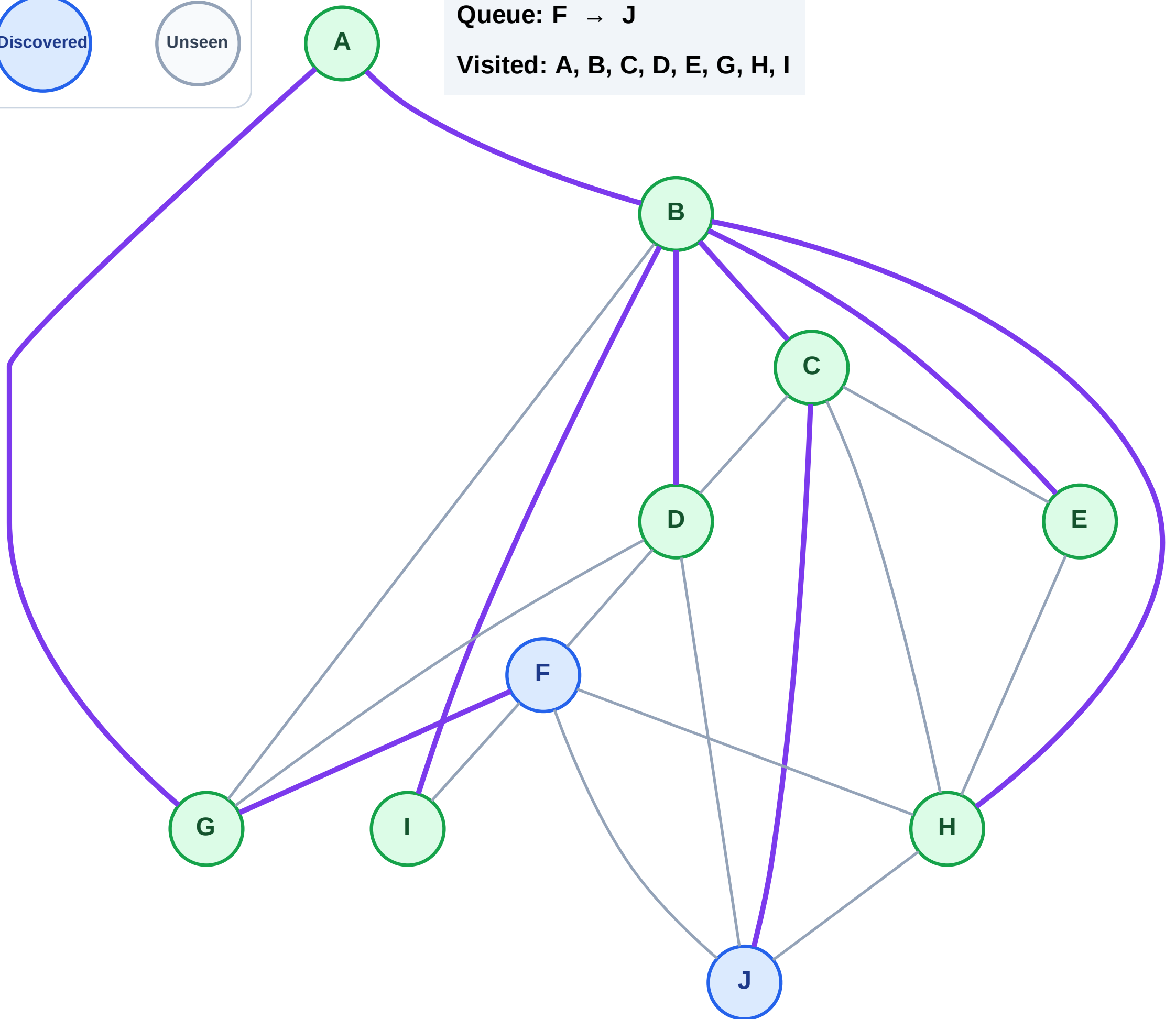
Step 17: No new neighbors from I

Legend



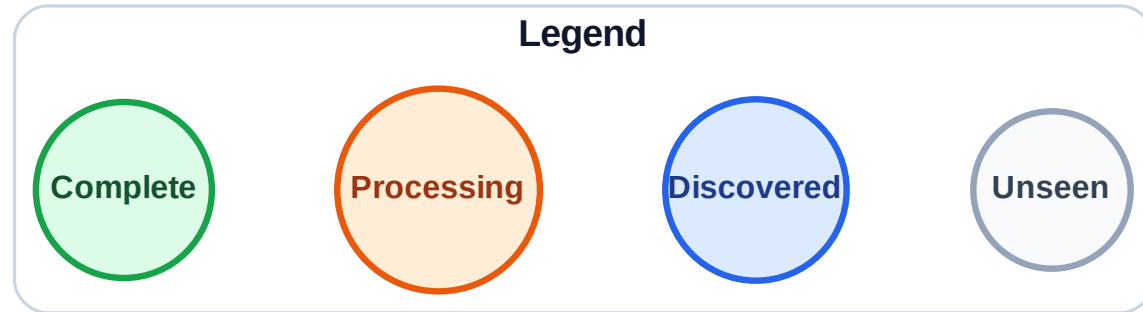
Queue: F → J

Visited: A, B, C, D, E, G, H, I



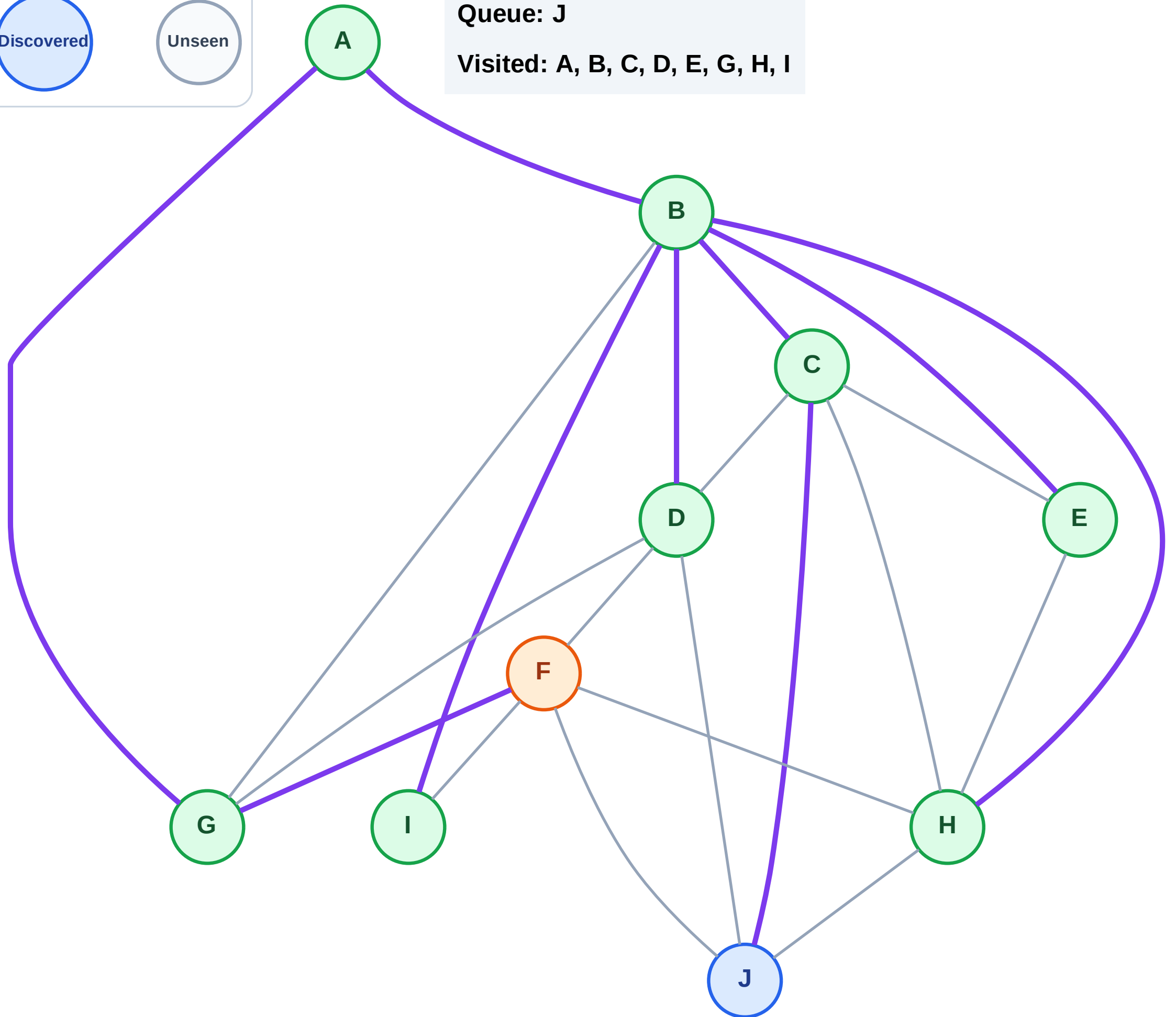
Step 18: Process node F

Step 18: Process node F



Queue: J

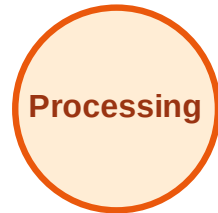
Visited: A, B, C, D, E, G, H, I



BFS Traversal

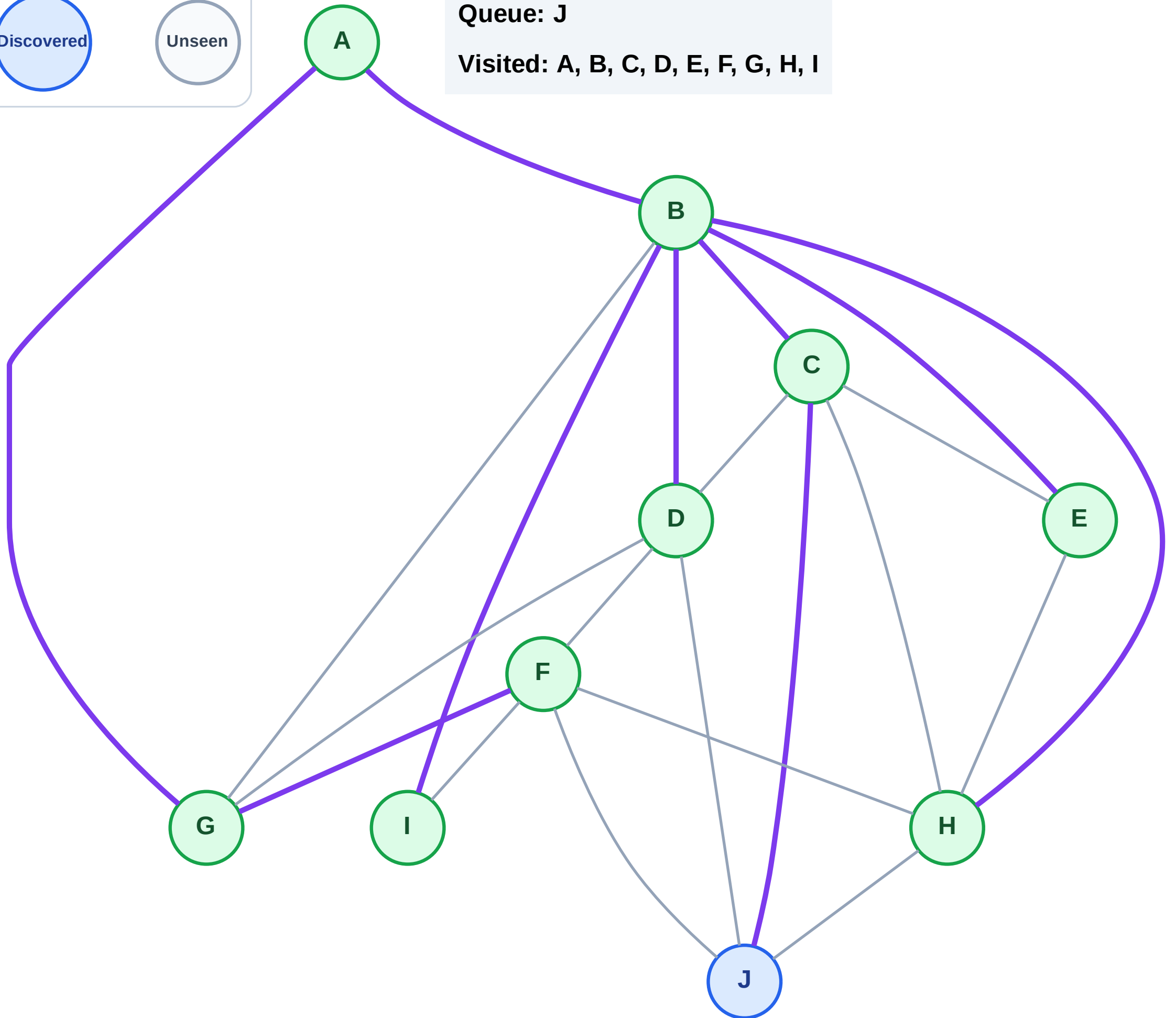
Step 19: No new neighbors from F

Legend



Queue: J

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

Step 20: Process node J

Legend

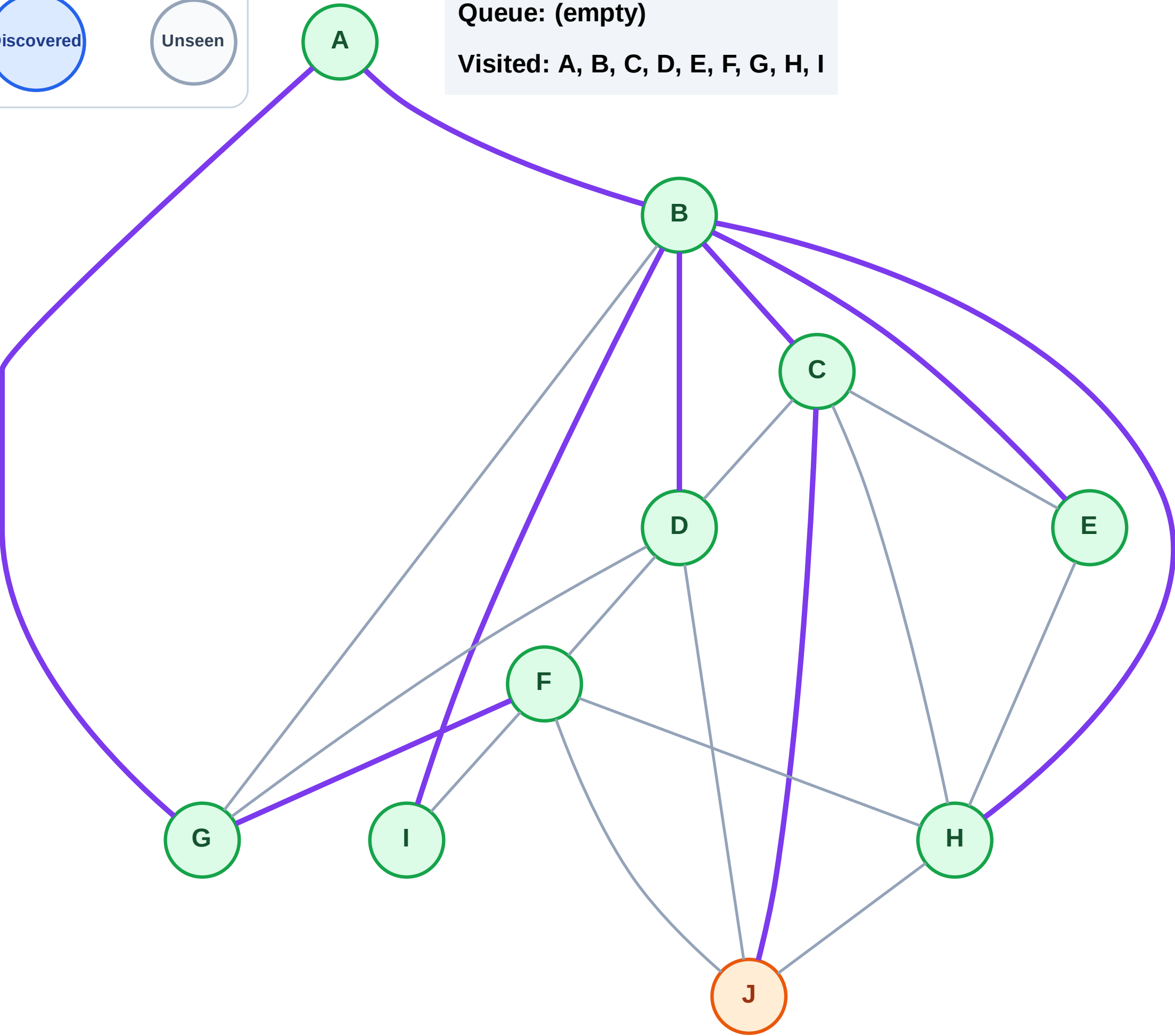
Complete

Processing

Discovered

Unseen

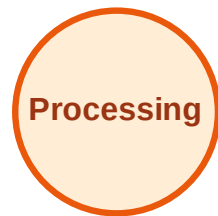
Queue: (empty)
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

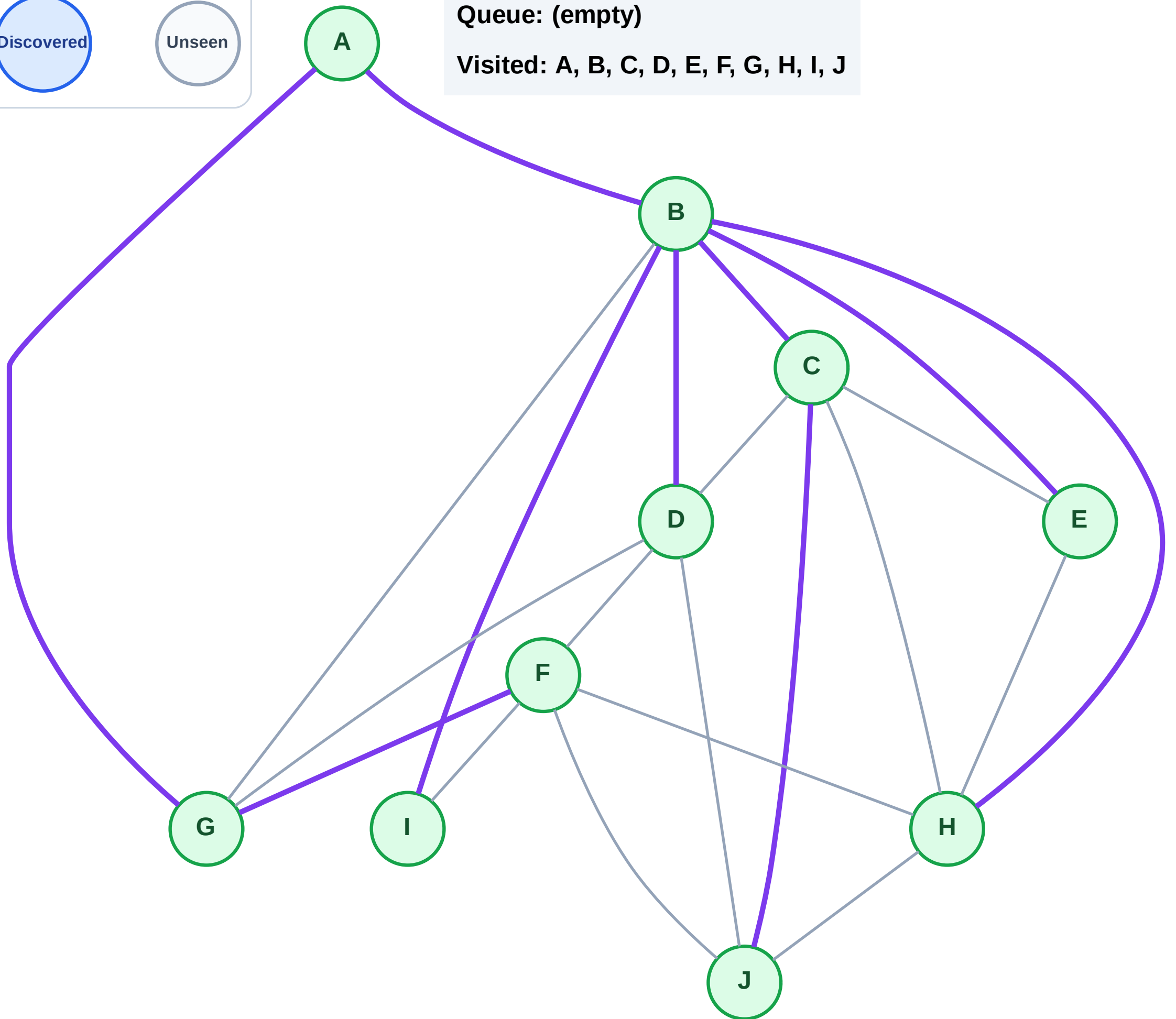
Step 21: No new neighbors from J

Legend



Queue: (empty)

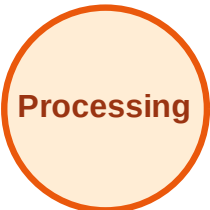
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

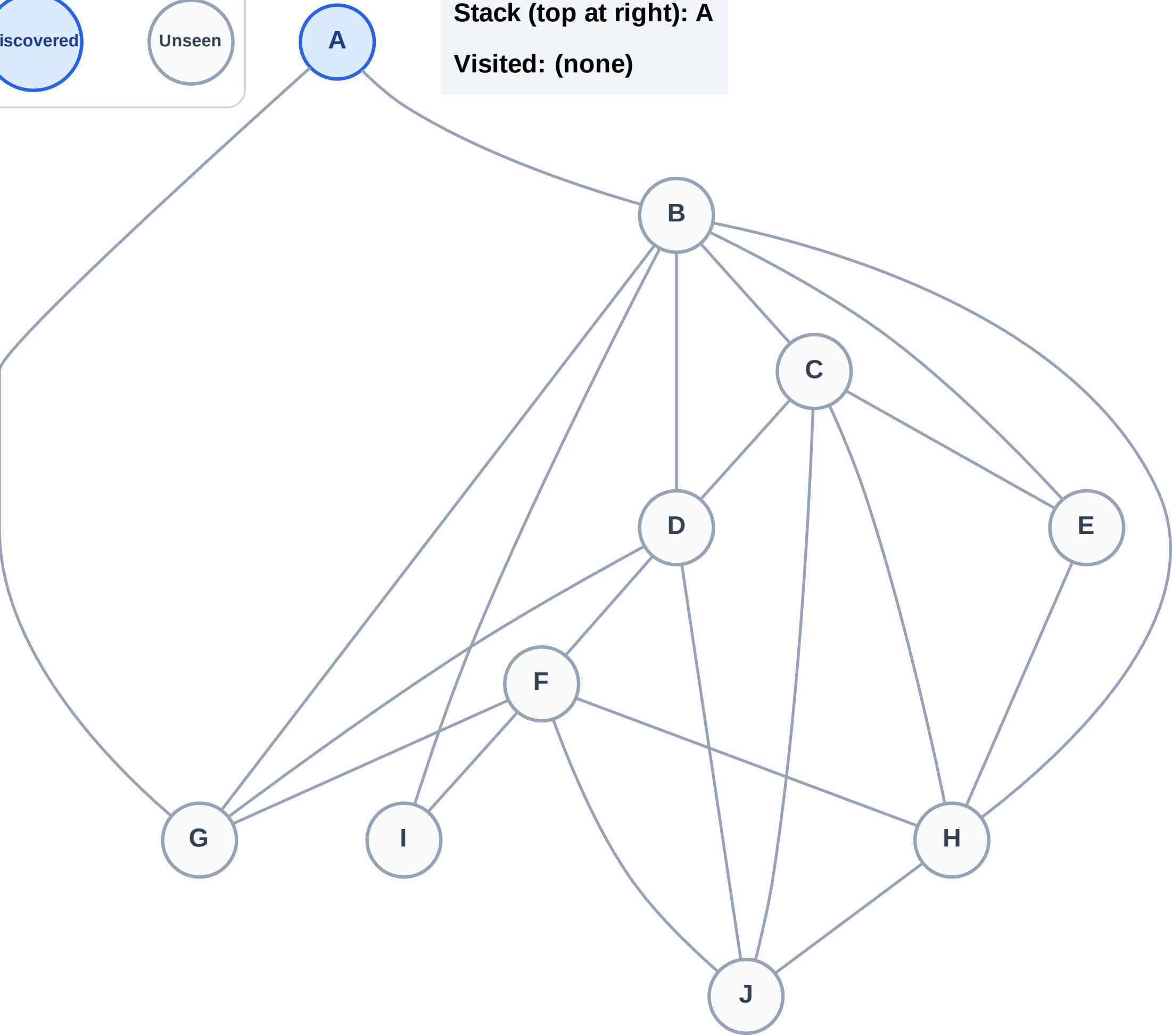
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

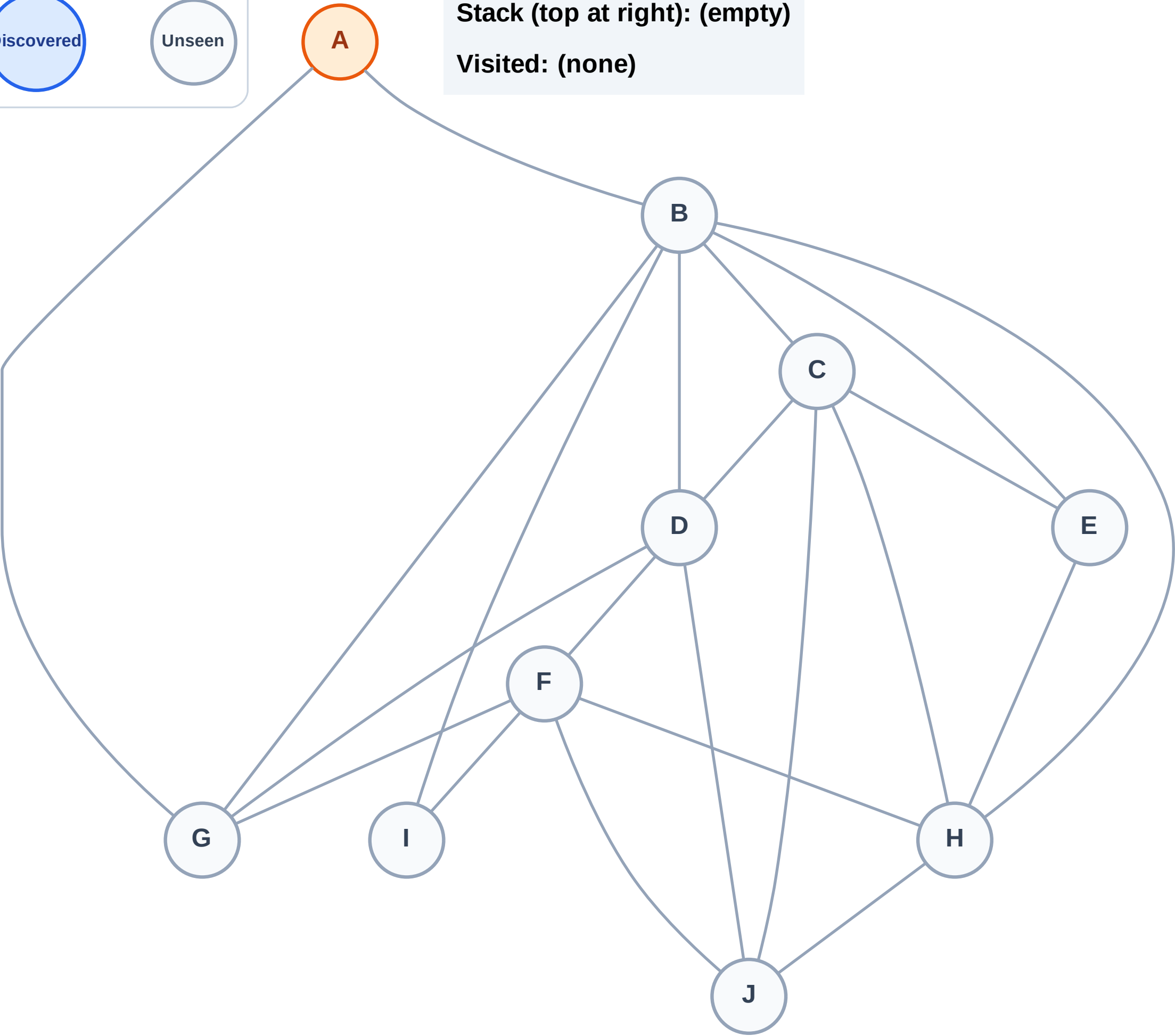
Complete

Processing

Discovered

Unseen

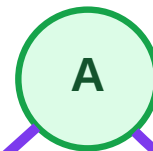
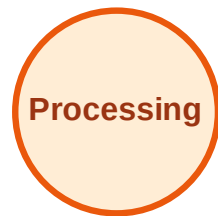
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

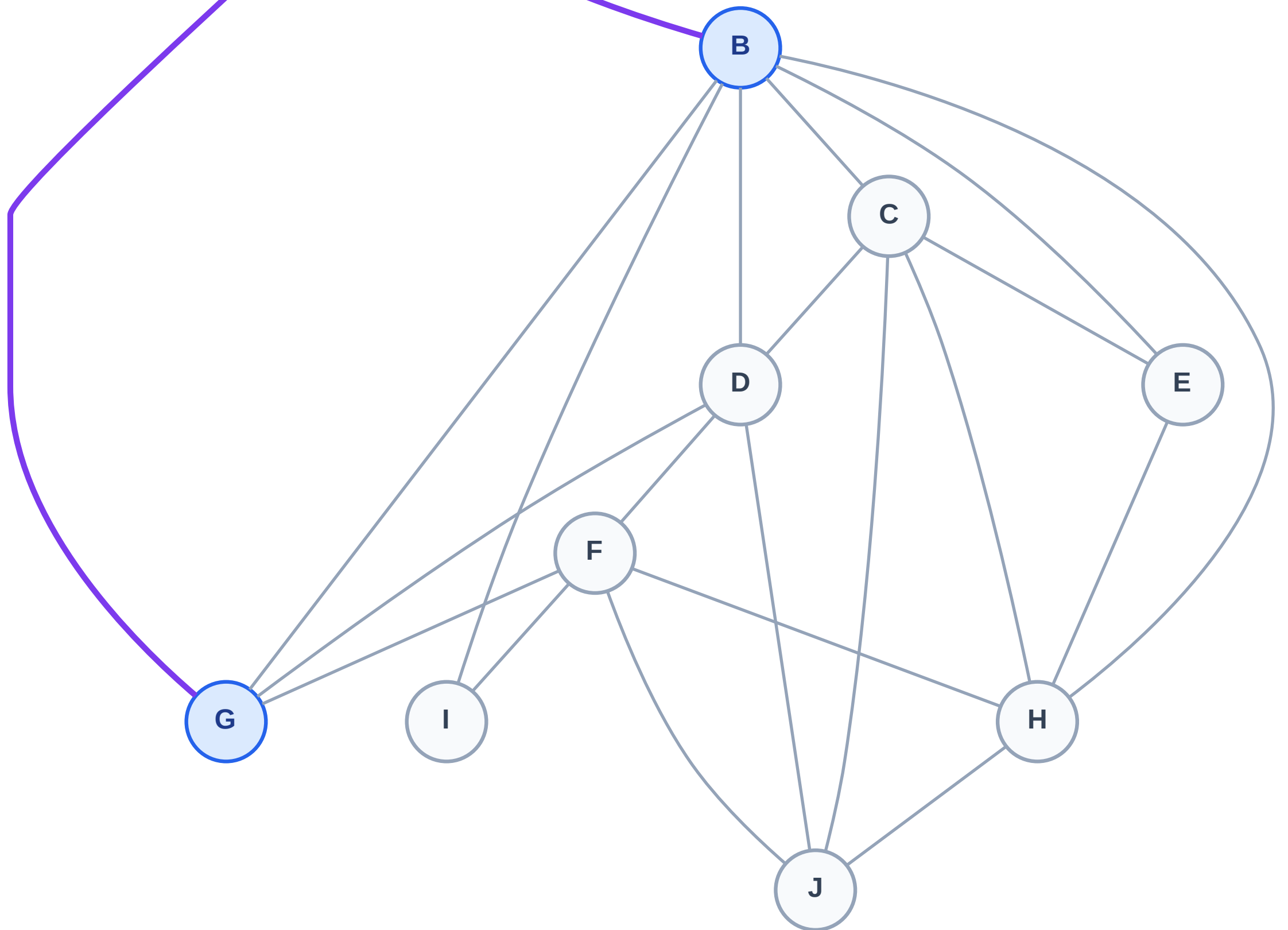
Step 3: Discover G, B from A

Legend



Stack (top at right): G | B

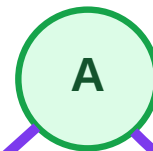
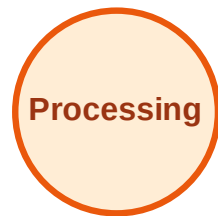
Visited: A



DFS Traversal

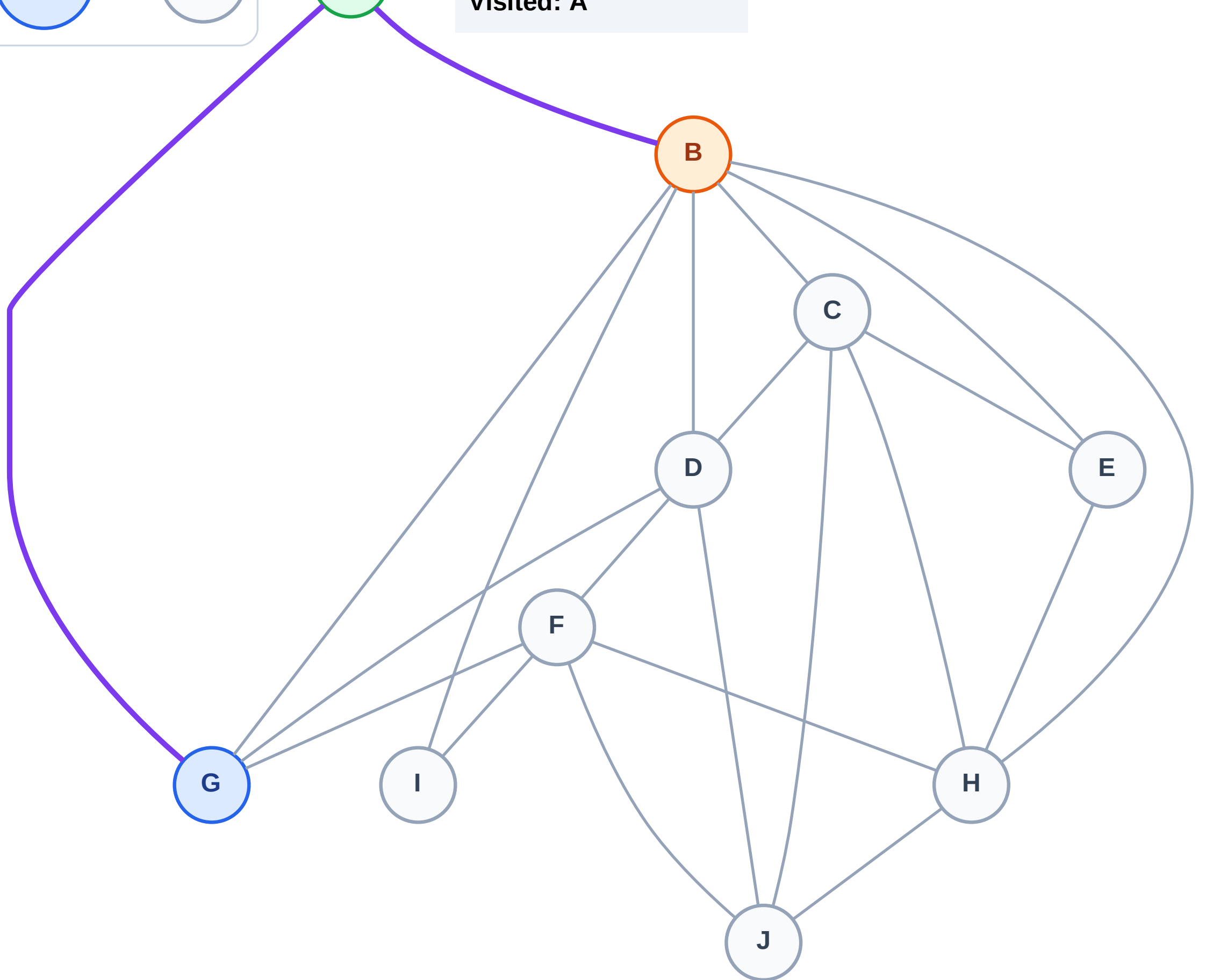
Step 4: Process node B

Legend



Stack (top at right): G

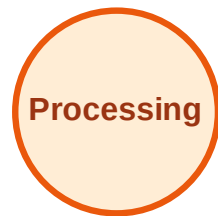
Visited: A



DFS Traversal

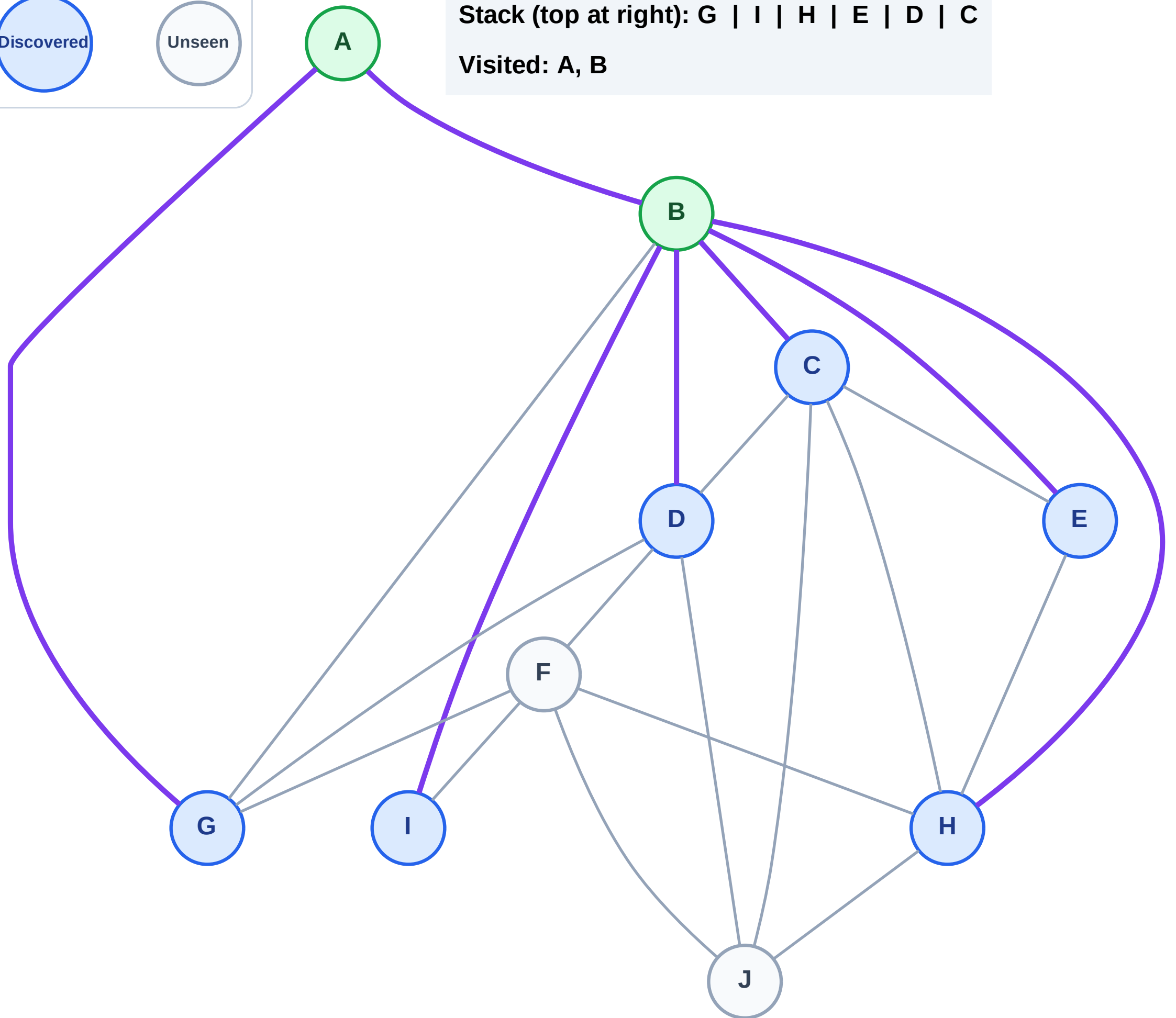
Step 5: Discover I, H, E, D, C from B

Legend



Stack (top at right): G | I | H | E | D | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

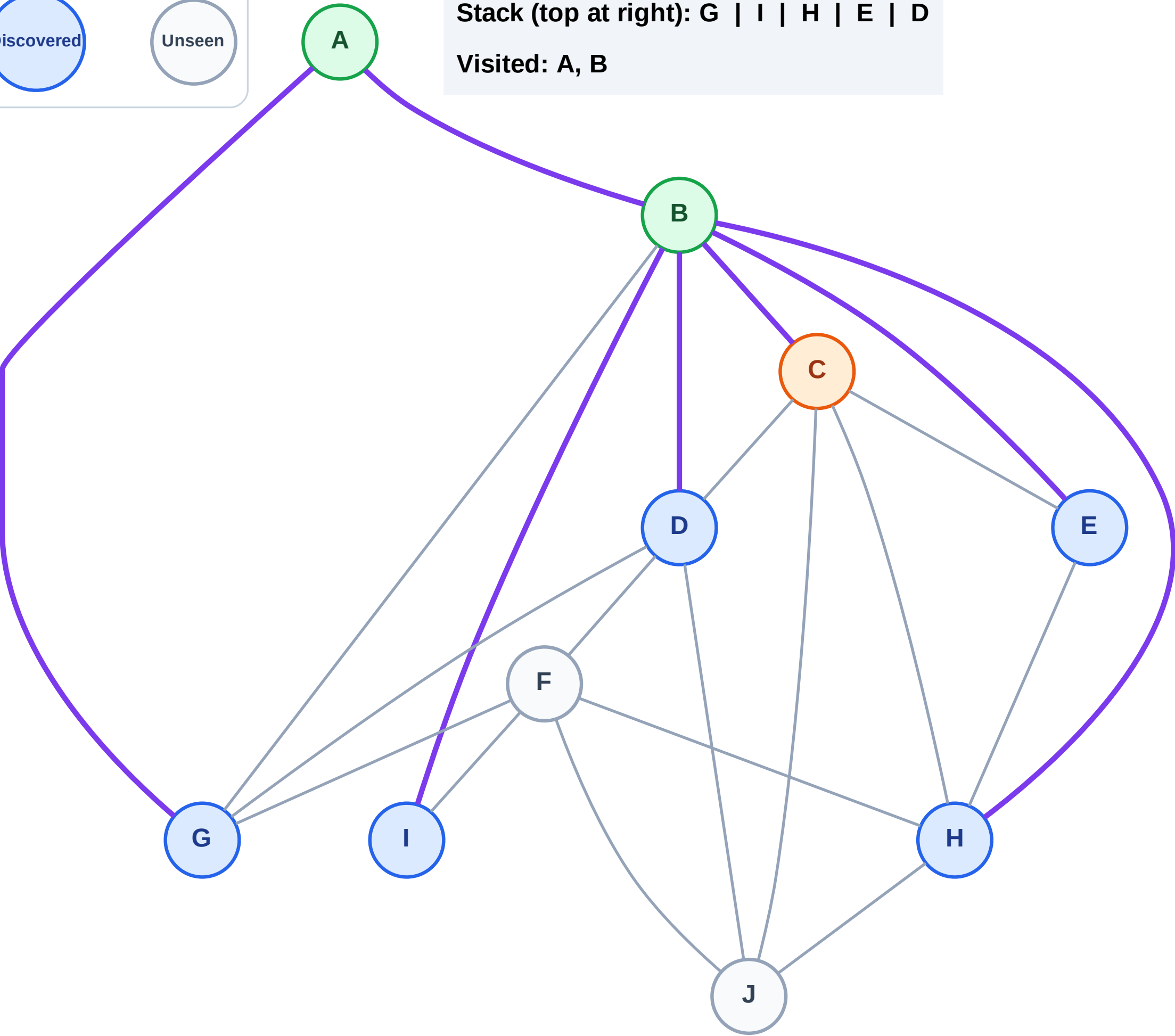
Processing

Discovered

Unseen

Stack (top at right): G | I | H | E | D

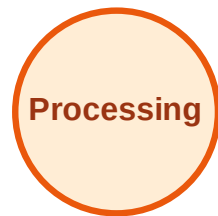
Visited: A, B



DFS Traversal

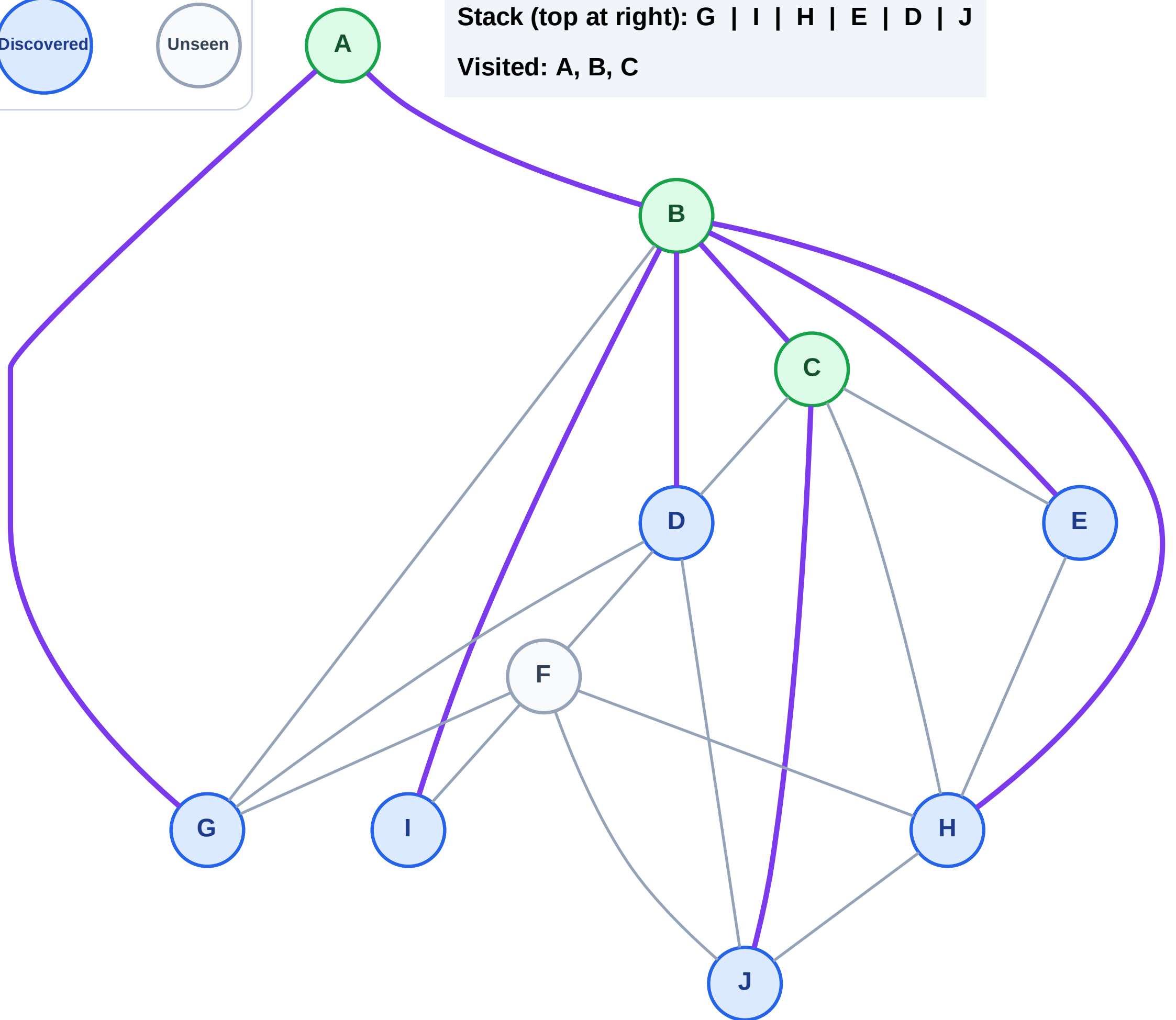
Step 7: Discover J from C

Legend



Stack (top at right): G | I | H | E | D | J

Visited: A, B, C



DFS Traversal

Step 8: Process node J

Legend

Complete

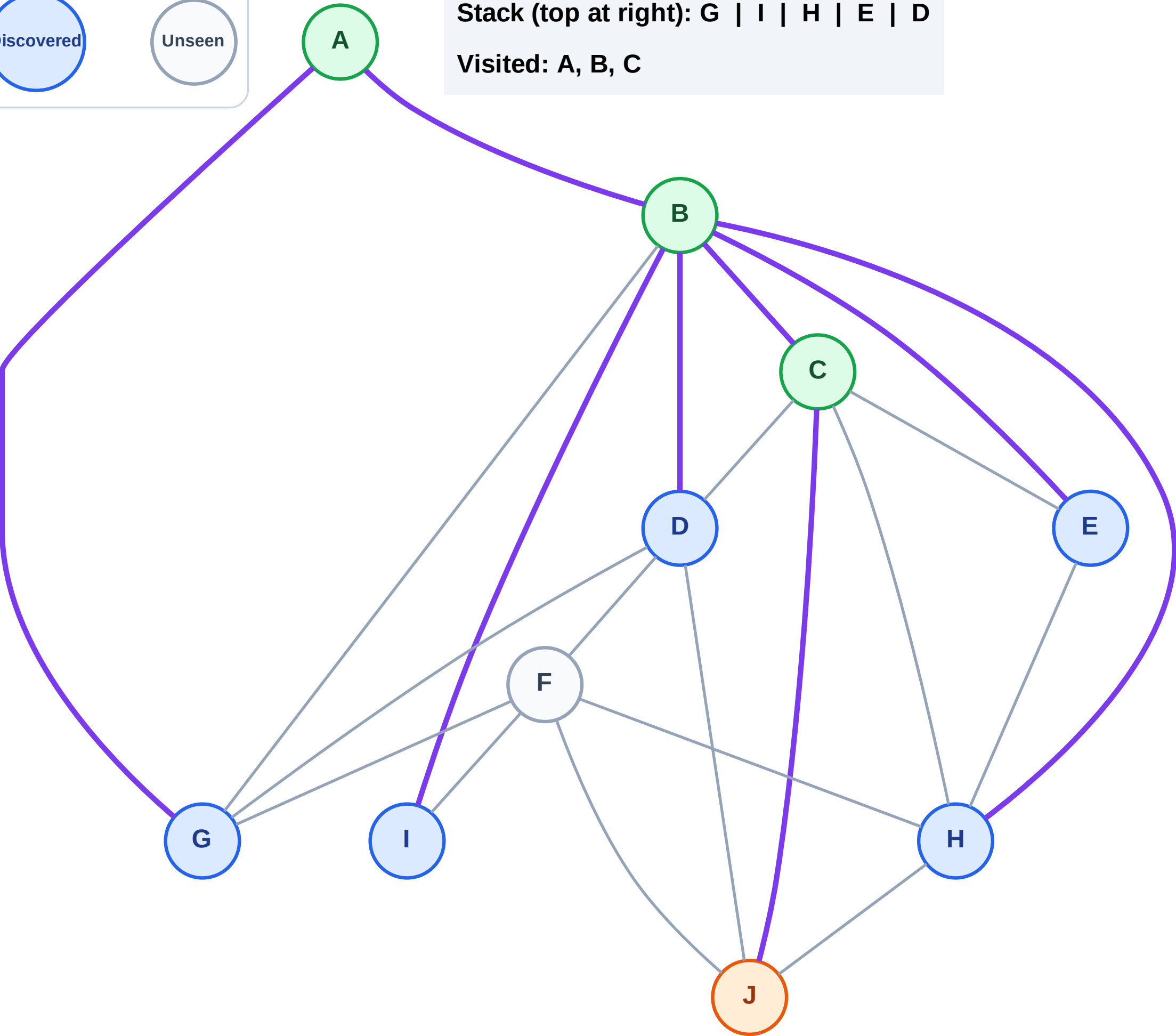
Processing

Discovered

Unseen

Stack (top at right): G | I | H | E | D

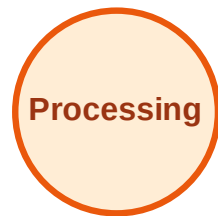
Visited: A, B, C



DFS Traversal

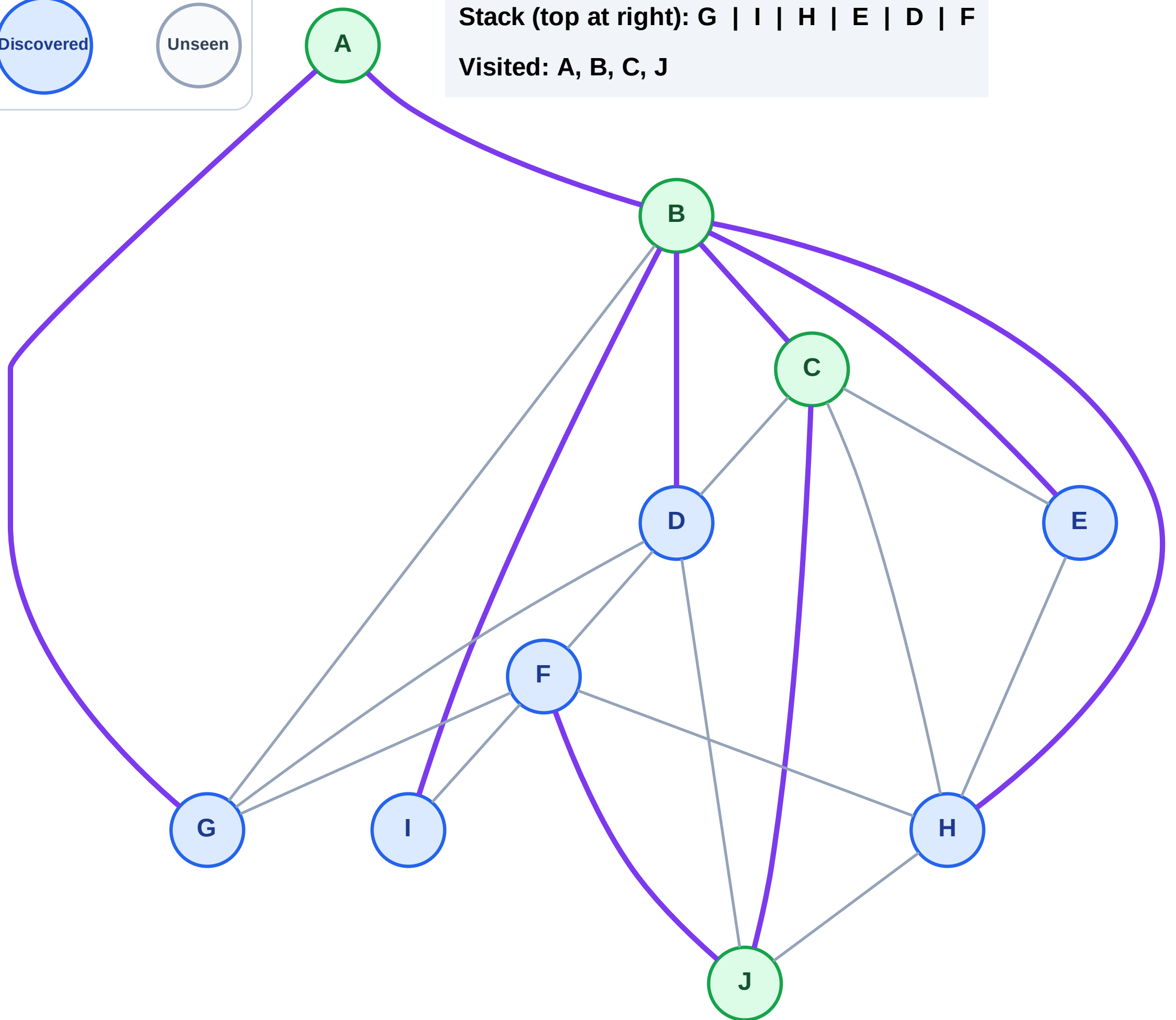
Step 9: Discover F from J

Legend



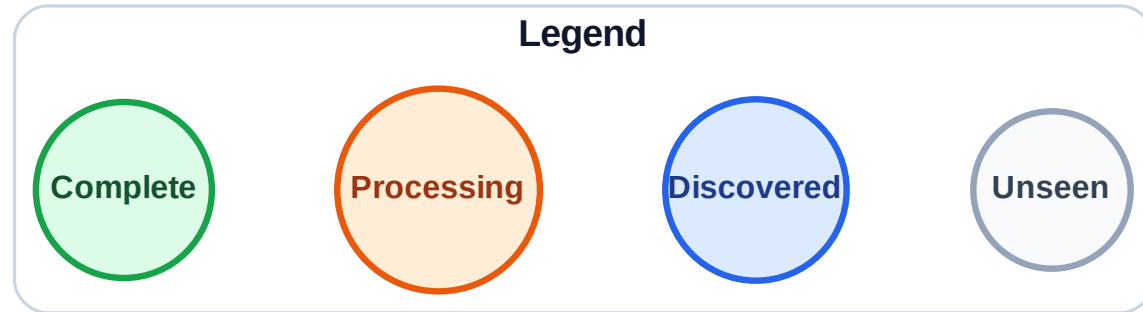
Stack (top at right): G | I | H | E | D | F

Visited: A, B, C, J



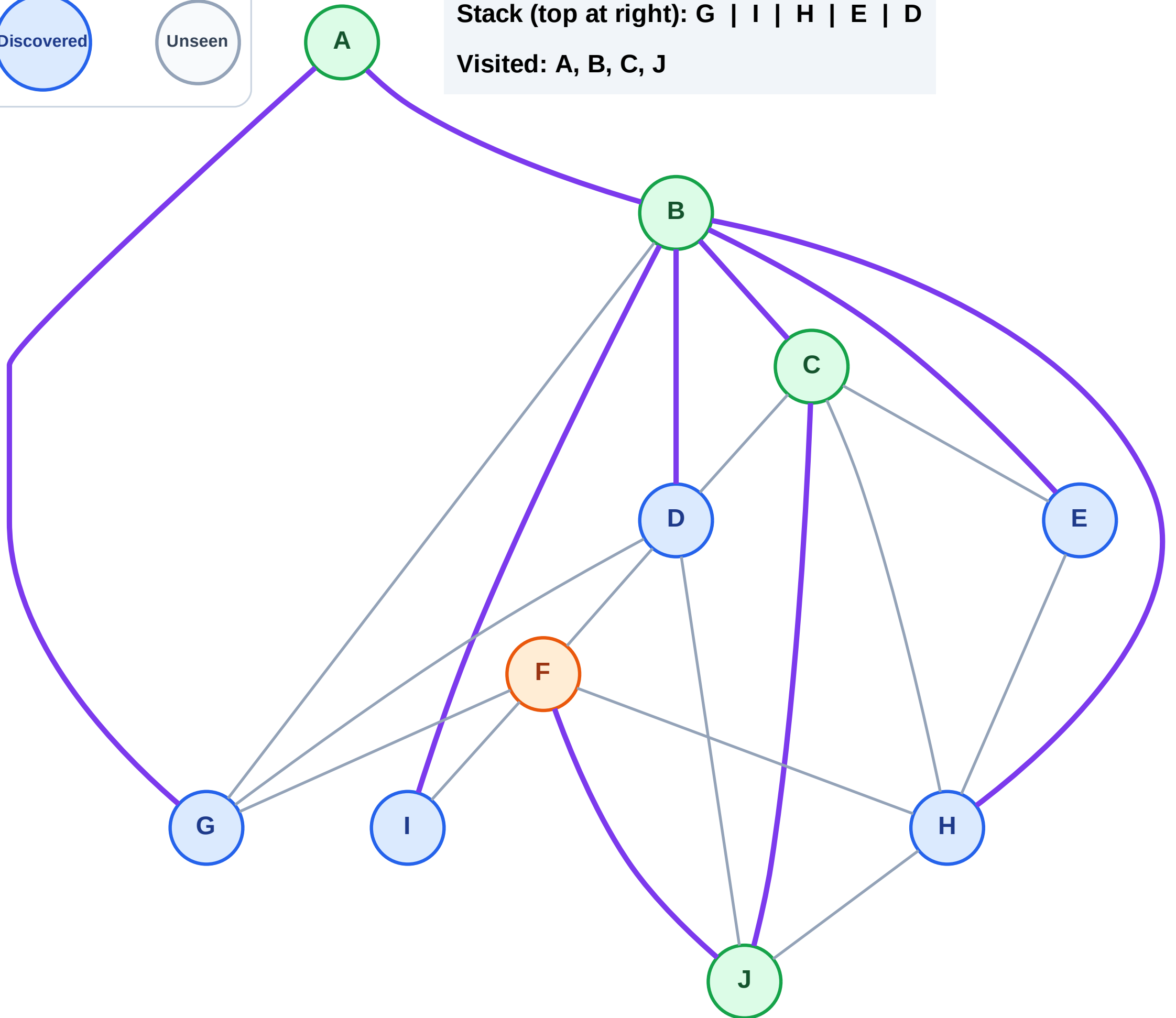
Step 10: Process node F

Step 10: Process node F



Stack (top at right): G | I | H | E | D

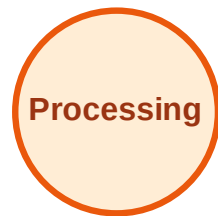
Visited: A, B, C, J



DFS Traversal

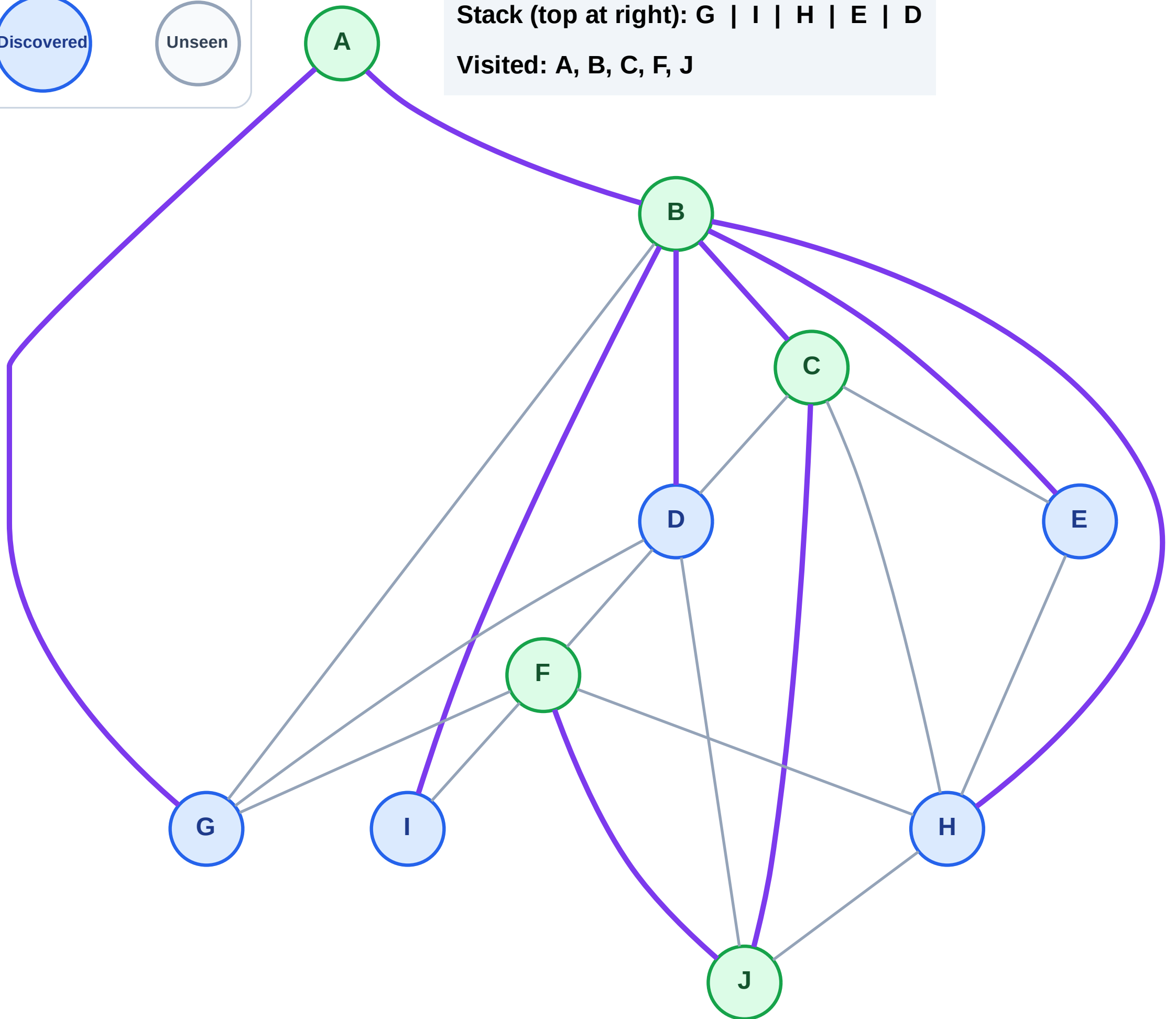
Step 11: No new neighbors from F

Legend



Stack (top at right): G | I | H | E | D

Visited: A, B, C, F, J



DFS Traversal

Step 12: Process node D

Legend

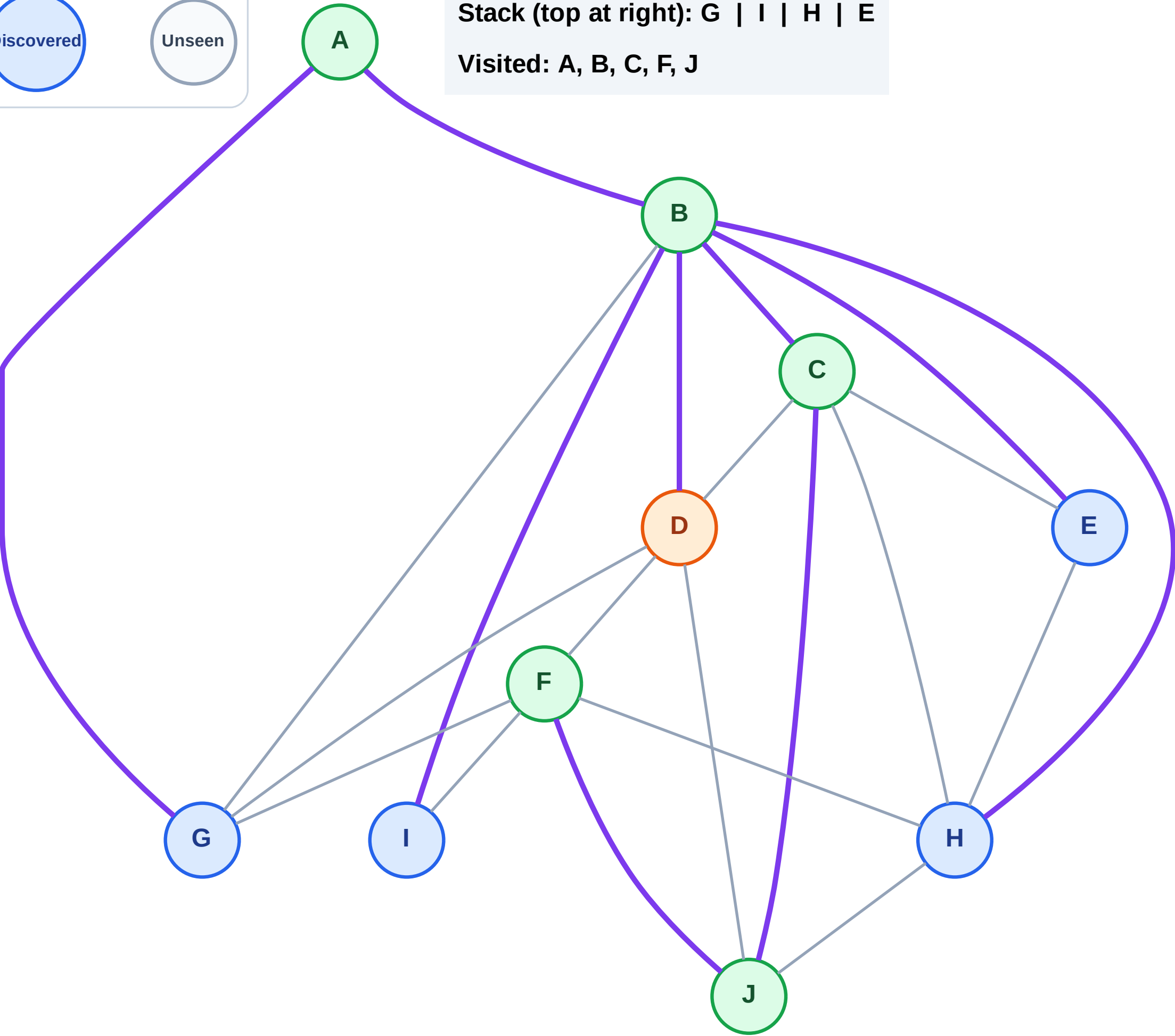
Complete

Processing

Discovered

Unseen

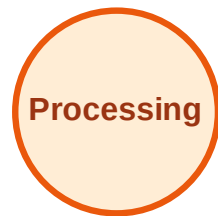
Stack (top at right): G | I | H | E
Visited: A, B, C, F, J



DFS Traversal

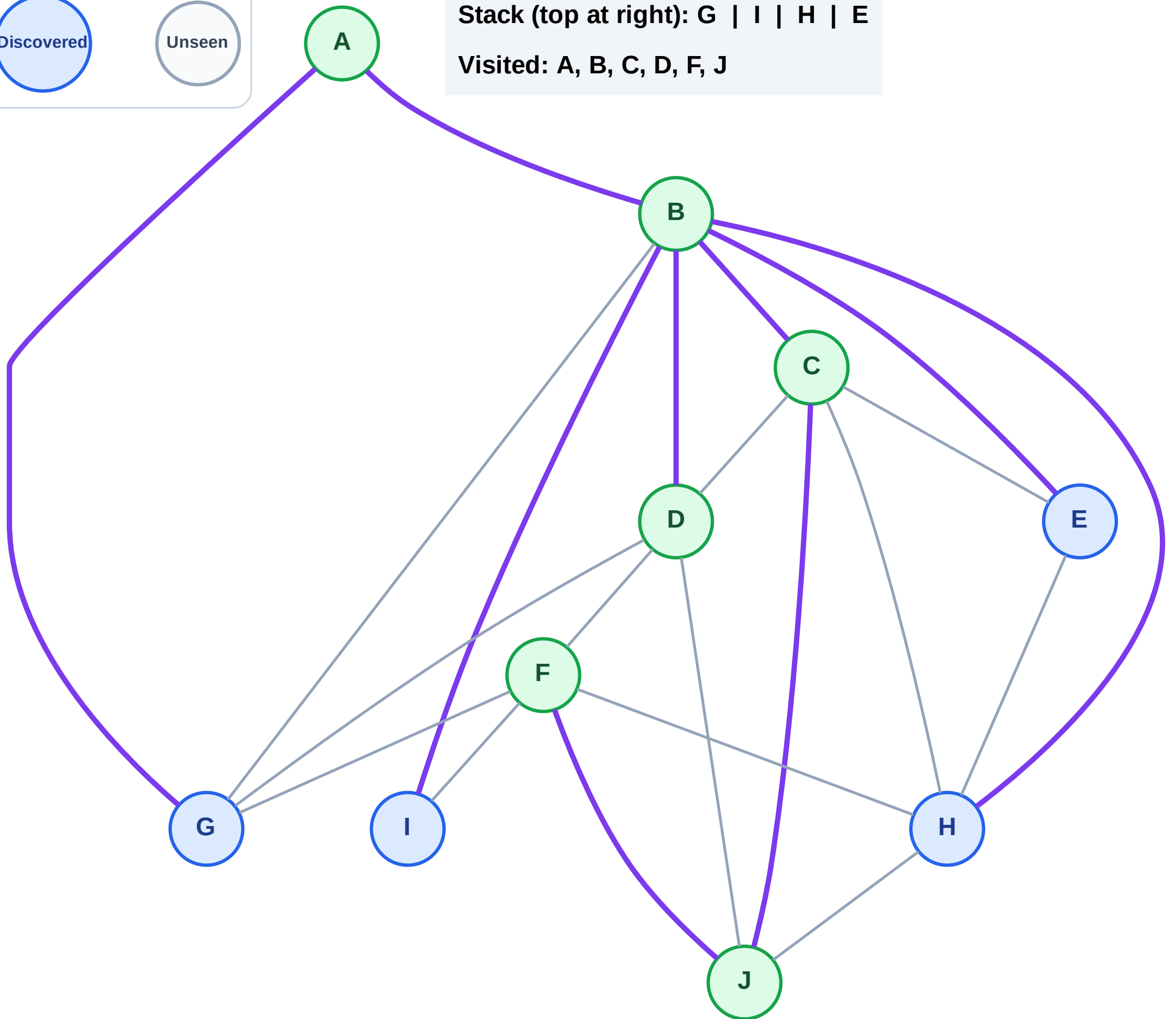
Step 13: No new neighbors from D

Legend



Stack (top at right): G | I | H | E

Visited: A, B, C, D, F, J



DFS Traversal

Step 14: Process node E

Legend

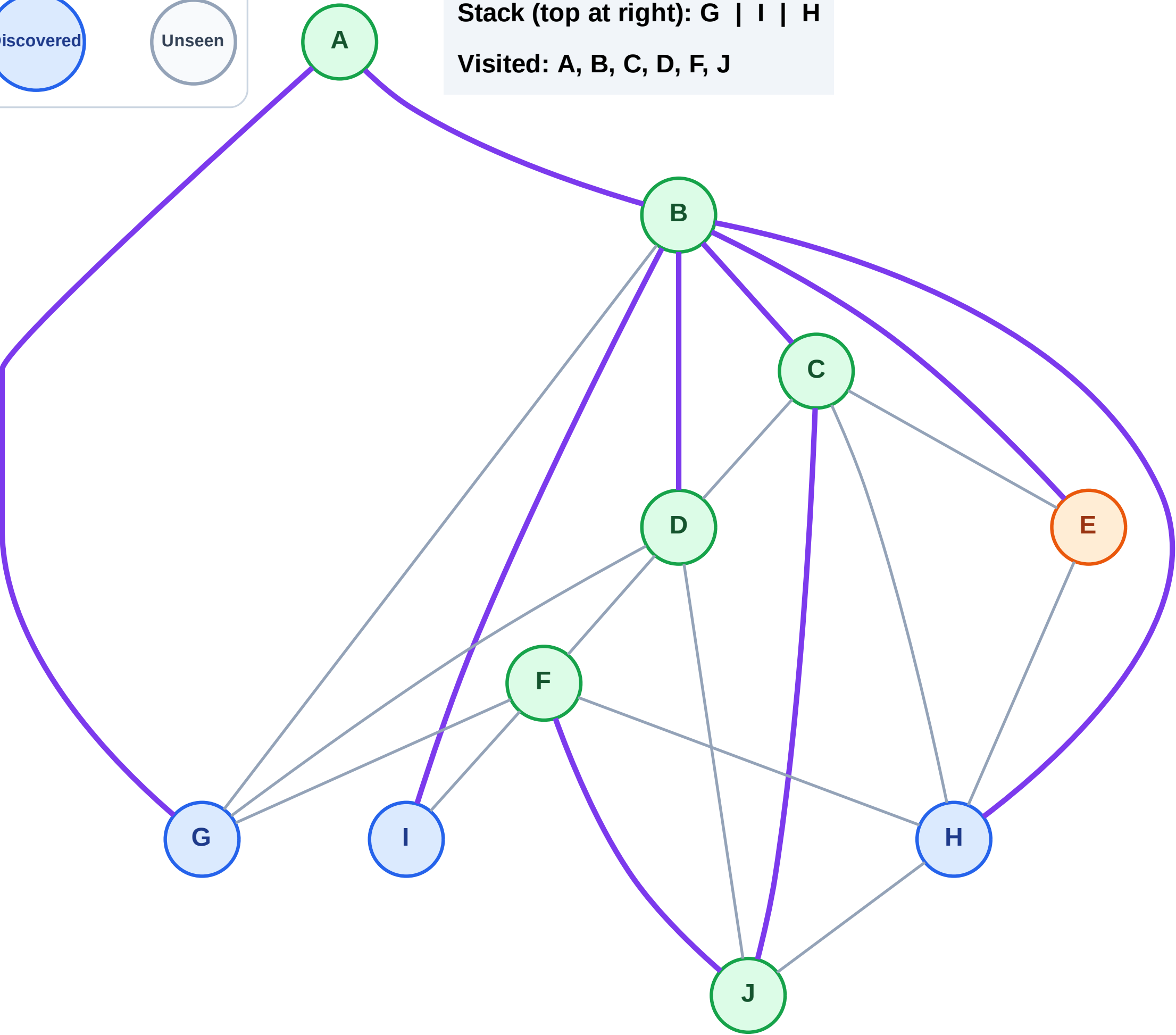
Complete

Processing

Discovered

Unseen

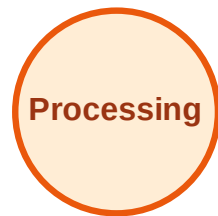
Stack (top at right): G | I | H
Visited: A, B, C, D, F, J



DFS Traversal

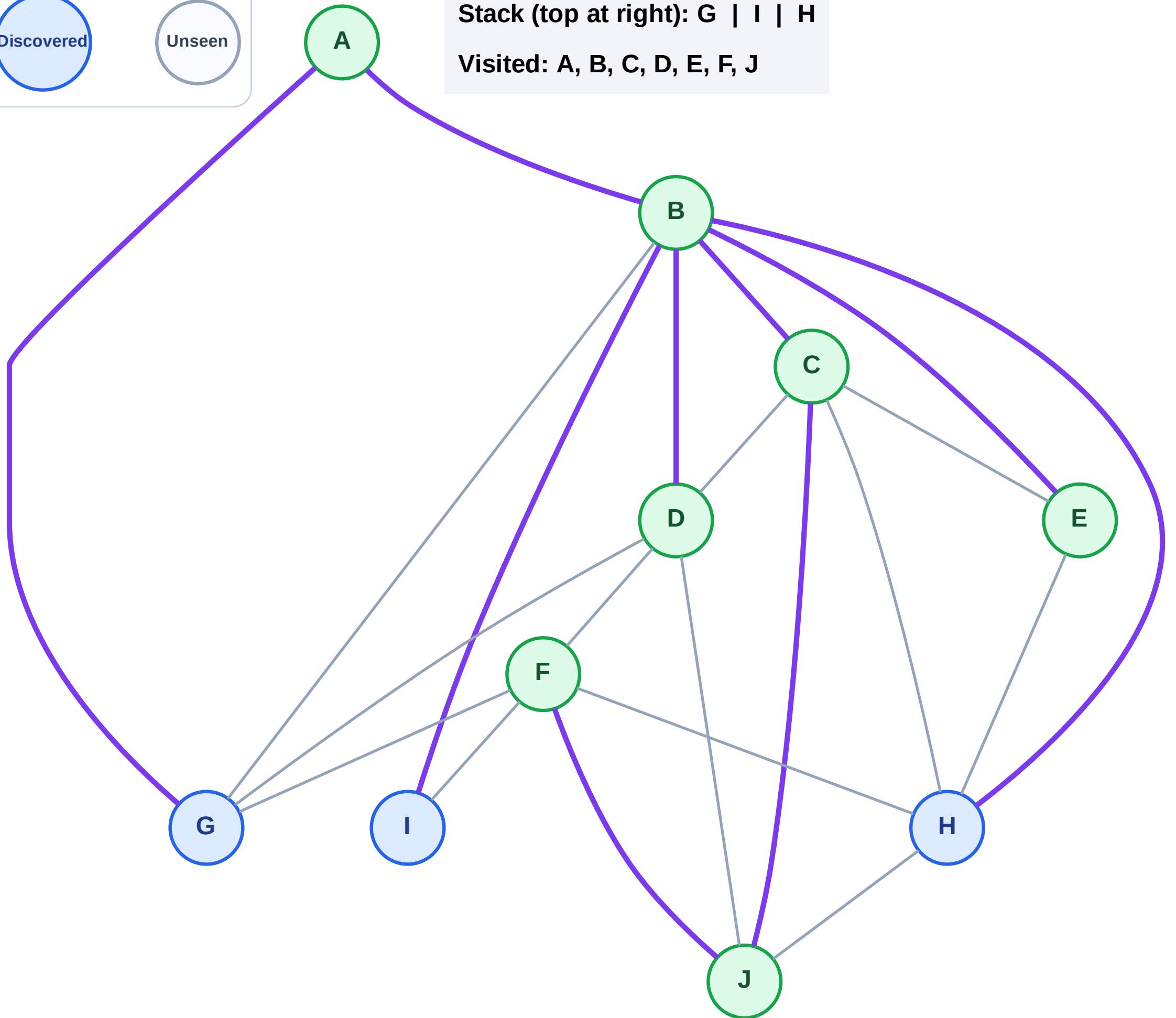
Step 15: No new neighbors from E

Legend



Stack (top at right): G | I | H

Visited: A, B, C, D, E, F, J



DFS Traversal

Step 16: Process node H

Legend

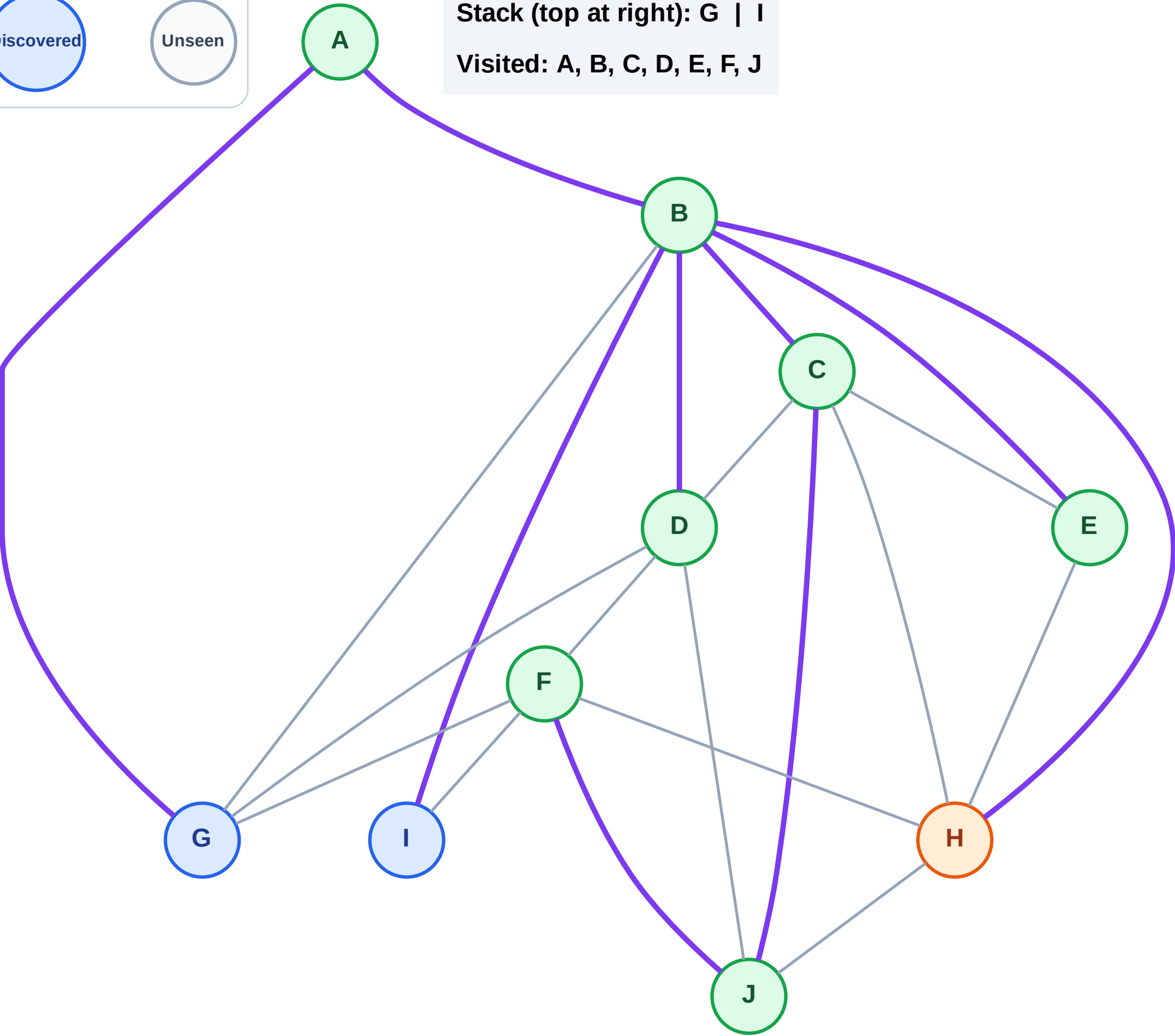
Complete

Processing

Discovered

Unseen

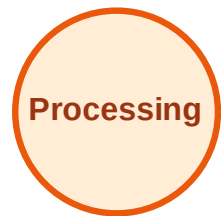
Stack (top at right): G | I
Visited: A, B, C, D, E, F, J



DFS Traversal

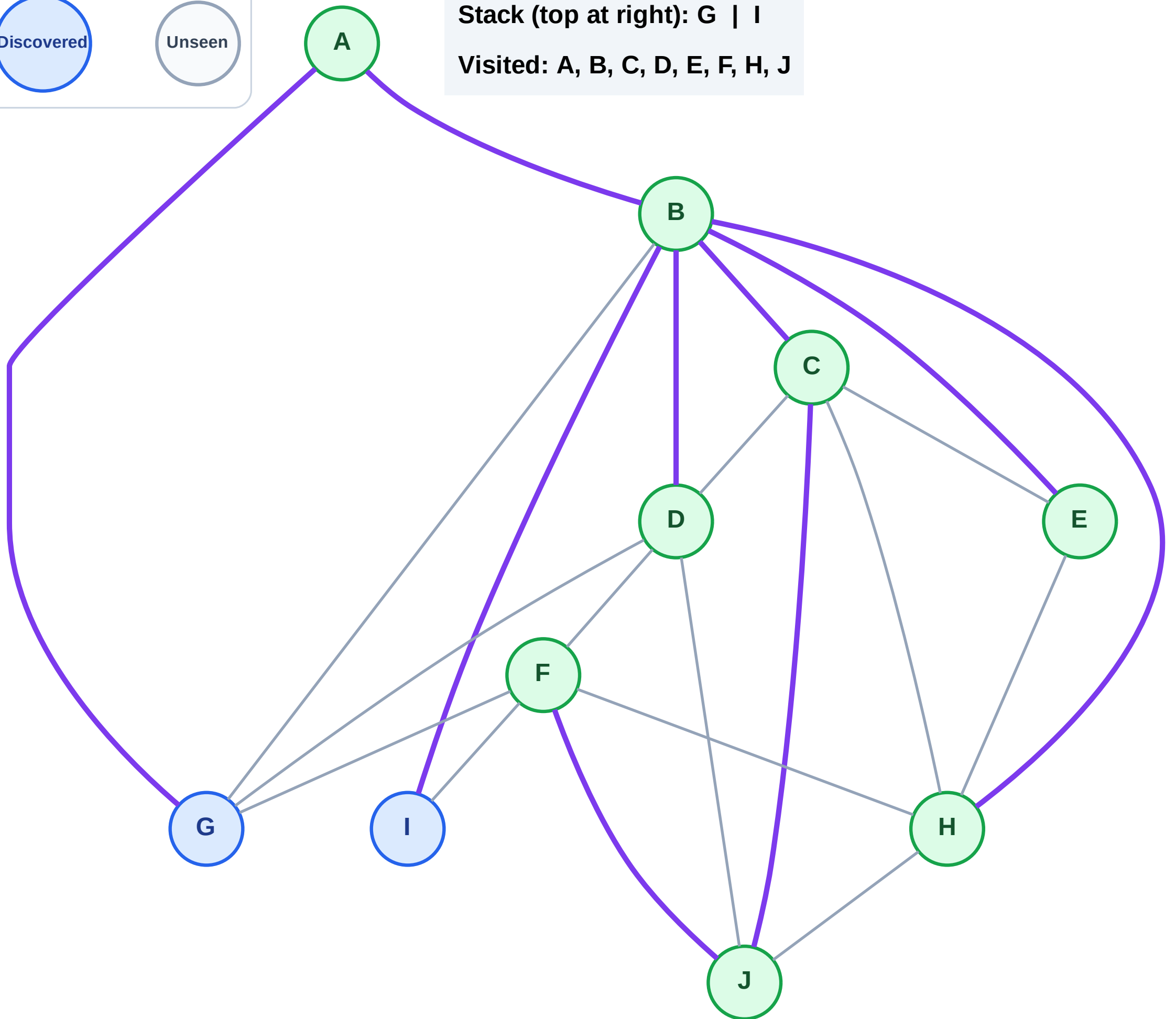
Step 17: No new neighbors from H

Legend



Stack (top at right): G | I

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node I

Legend

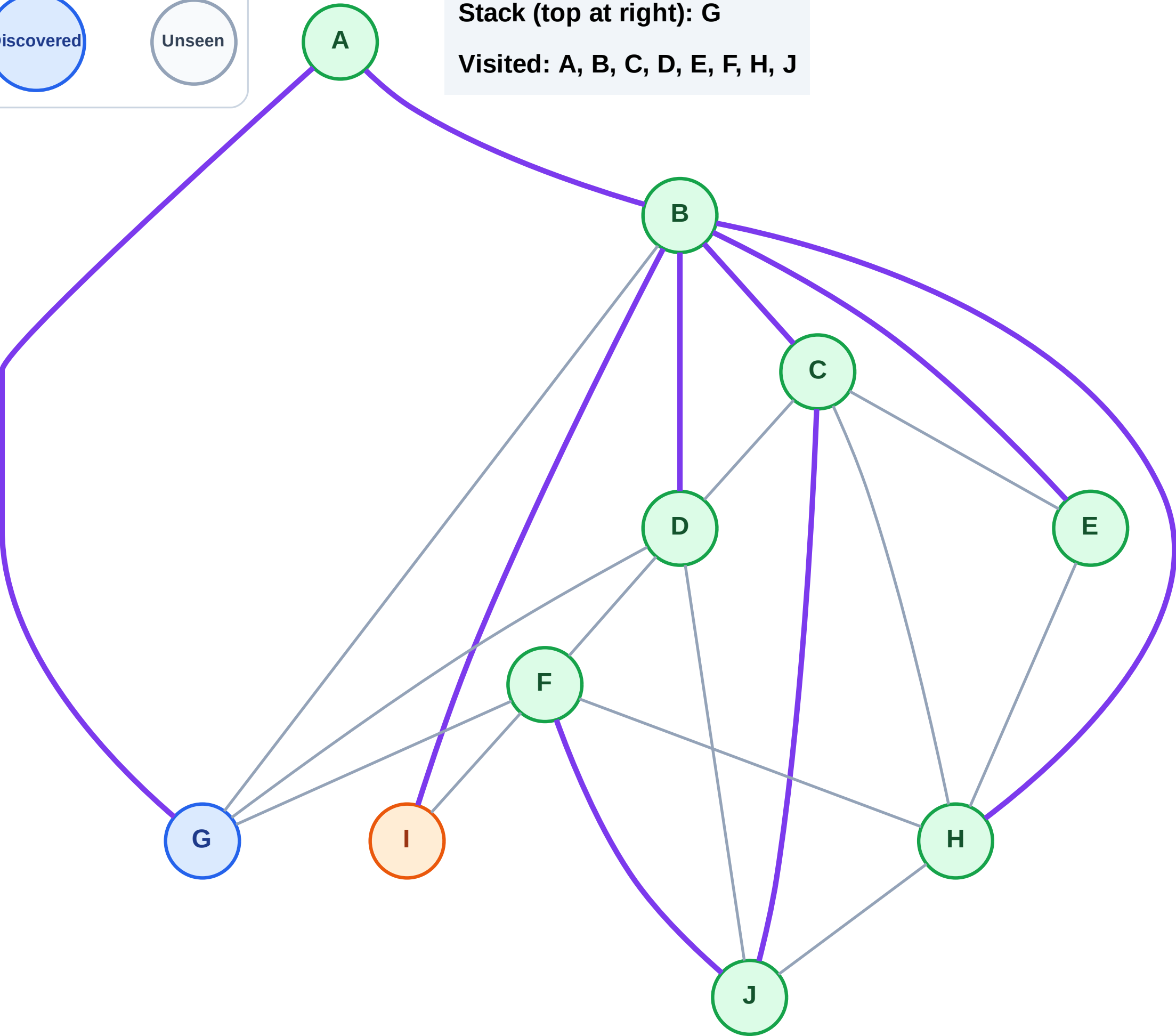
Complete

Processing

Discovered

Unseen

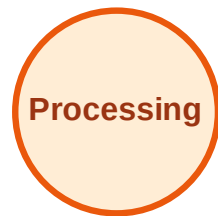
Stack (top at right): G
Visited: A, B, C, D, E, F, H, J



DFS Traversal

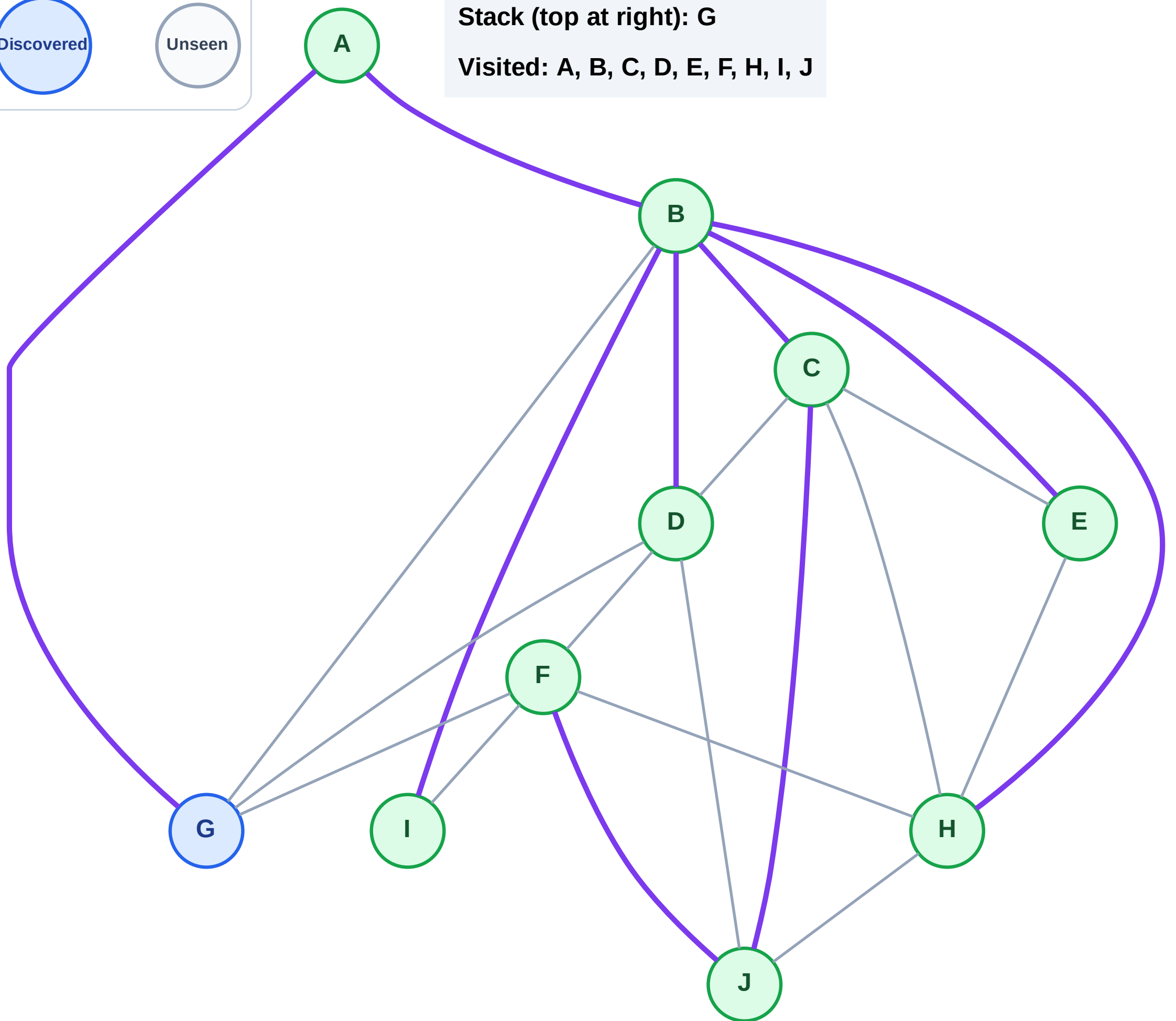
Step 19: No new neighbors from I

Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 20: Process node G

Legend

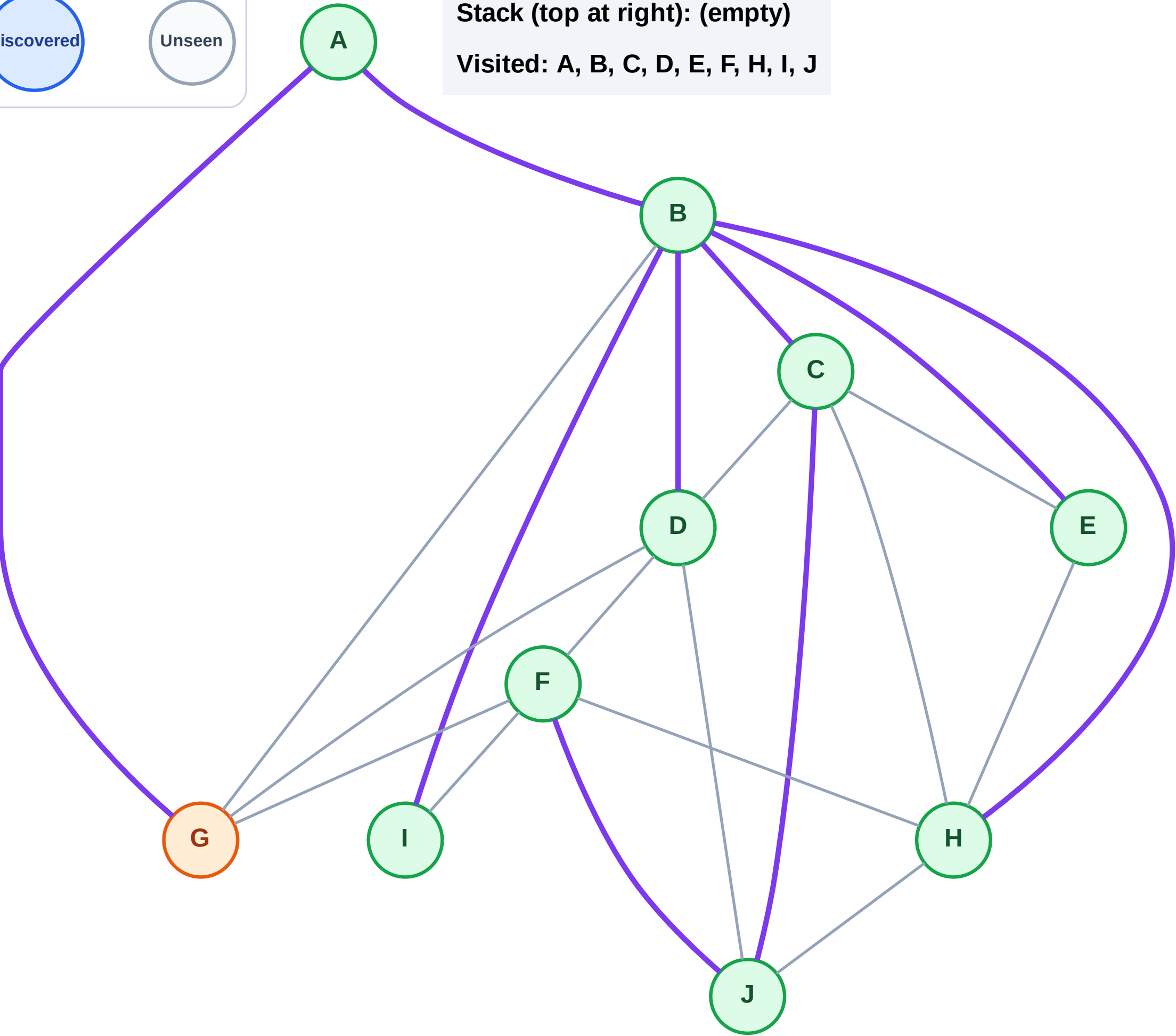
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, H, I, J



Step 21: No new neighbors from G

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

